A Quality Model-based Approach for Measuring User Interface Aesthetics with Grace

MATHIEU ZEN, Université catholique de Louvain, Belgium
NICOLAS BURNY, Université catholique de Louvain, Belgium
JEAN VANDERDONCKT, Université catholique de Louvain, Belgium

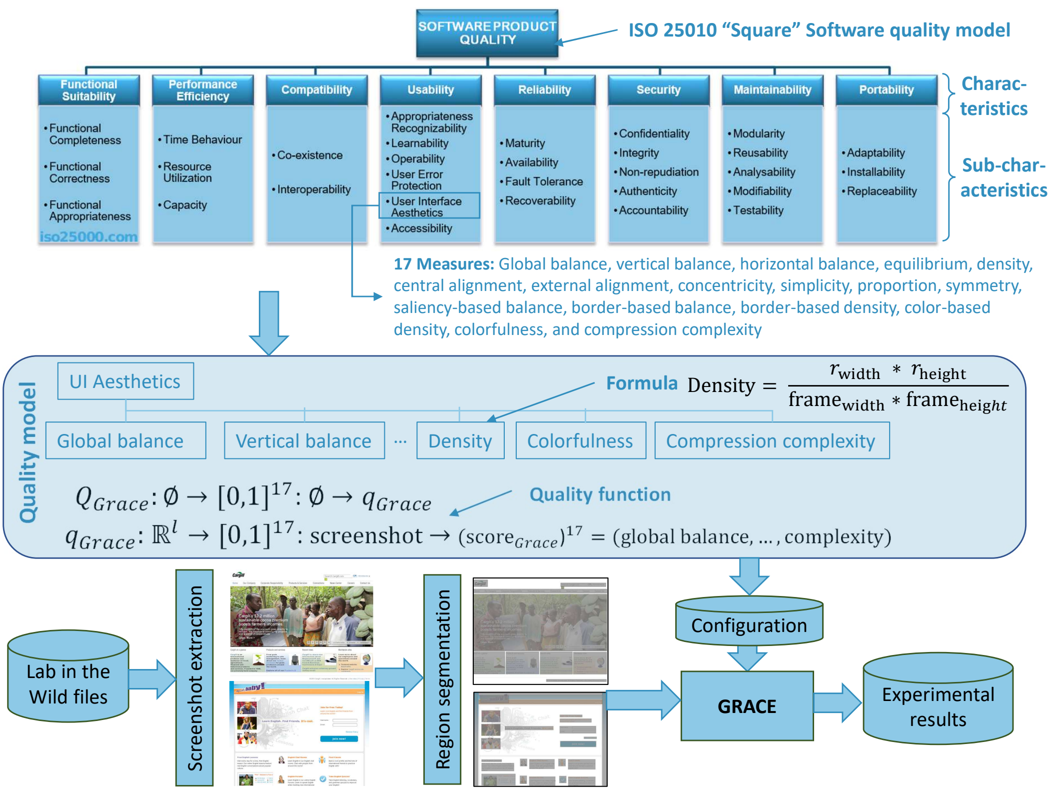

Fig. 1. Overview of the quality model-based approach supported by Grace and involved concepts.

Authors' addresses: Mathieu Zen, Université catholique de Louvain, Exploitation des produits et services du système d'information (SIPS), Pierre & Marie Curie, rue du Compas 1, 1348, Louvain-la-Neuve, Belgium, mathieu.zen@uclouvain.be; Nicolas Burny, Université catholique de Louvain, Louvain Research Institute in Management and Organizations, Place des Doyens, 1, Louvain-la-Neuve, 1348, Belgium, nicolas.burny@uclouvain.be; Jean Vanderdonckt, Université catholique de Louvain, Louvain Research Institute in Management and Organizations, Place des Doyens, 1, Louvain-la-Neuve, 1348, Belgium, jean.vanderdonckt@uclouvain.be.

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User interface aesthetics, a particular sub-characteristic of the ISO 25010 software quality model, is correlated to the perceived or actual usability of a graphical user interface, its user experience, and trust. While many measures, such as balance, symmetry, proportion, alignment, regularity, and simplicity, can be computed, no consensus exists today on which measure to adopt, which formula to compute for each measure, and which interpretation to give for each computed formula. To accommodate these variations and to make the assessment explicit and interpretable, we present a quality model-based approach for measuring aesthetics that defines a quality function in terms of the formula of the measures, their weights, their composition, and their overall computation. This quality model is transformed into a configuration used by Grace, a web application developed for this purpose. When the measures, their formula, their weights, or the quality function change, the quality model changes, which is re-computed to compare it with any other model, thus making the measurement process explicit and interpretable. We apply this approach to a large dataset "Lab in the Wild" to investigate the correlations between aesthetic measures and perceived visual complexity and colorfulness. We discuss limitations through threats to validity.

CCS Concepts: • **Human-centered computing → Graphical user interfaces**; **Web-based interaction**; **HCI design and evaluation methods**; *Usability testing*; Systems and tools for interaction design.

Additional Key Words and Phrases: Aesthetics, Graphical User Interfaces, Software measurement, User interface evaluation, Usability engineering, Visual measures, Visual techniques.

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1 INTRODUCTION

The ISO standard 25010 "Systems and Software Quality Requirements and Evaluation (SQuaRE)" [41] characterizes software quality according to eight characteristics, which are broken down into sub-characteristics (Fig. 1). For example, the "Usability" characteristic is broken down into sub-characteristic, such as "Learnability" or "User interface aesthetics", the latter being defined as the "degree to which a user interface enables pleasing and satisfying interaction for the user" [41].

Assuming the importance, interest, and validity of such a standard [11, 12], we wonder how to put it into practice. Understanding how to obtain an operational definition of this standard is essential [28]. Although it could also be questioned (see Chapter 2 on the legitimacy of quantifying aesthetics [44]), a common attempt at operationalization is to adopt a measurable approach, *i.e.*, by defining a set of measures that can be calculated, estimated or approximated. To evaluate a (sub-)characteristic is to compute a value of these measures through a formula and to interpret the result, either for certification or to position the user interface (UI) in relation to others.

Adopting such a measurable approach can be defined in terms of measurement concepts, such as [55]: metric, measure, measurement, scale, unit, measurement method. However, if we consider the "UI Aesthetics", which is already intrinsically subjective and therefore difficult to objectify, even through inter-subjectivity [103], there is a wide range of measures with, each time, one or more different formulas and one or more interpretations that are sometimes difficult to reproduce correctly and that do not satisfy consistency. To evaluate aesthetics, there are several tendencies.

A **first**, classical tendency is to express aesthetics in the form of a series of visual techniques [98] borrowed from visual design [14, 27], such as balance, symmetry (assuming that these techniques are actually related to aesthetics [45]), and to associate a calculable formula [58, 71], preferably computable via a software [72], and an effective interpretation [59]. Otherwise, the approach is not really measurable and interpretable. For this purpose, a model and an evaluation method are required. For example, Abbasi et al. [1] modeled and evaluated "UI aesthetics" according to two methods: a breakdown into (sub-)characteristics (*e.g.*, "Visual appeal" is decomposed into "Object clarity", "Object size adequacy",...) is evaluated according to an expert inspection method and 7-point Likert scales are evaluated through a questionnaire. The final ratings were almost the same for two UIs [1], but the models were partially explicit (how individual scores are aggregated into the final rating is not clear), not really objective (both methods resulted in subjective ratings) and not computable (only a manual evaluation was performed). Therefore, they were not easily interpretable. Conversely, several software exists that perform this kind of evaluation explicitly and objectively (a real quality model is underlying), but it is rarely flexible. For example, GUIEVALUATOR [2] computes the complexity of the user interface, which is correlated to human assessment, but the underlying model is not visible and cannot be modified to investigate another quality model. Similarly, SLC [3] computes a cohesion metric to predict usability of the UI, but the model is not explicit, not visible, and not flexible. In most cases, the underlying quality model is inflexible, which prevents the practitioner or the researcher from creating another model and comparing it.

A **second** tendency is to apply machine learning (*e.g.*, a convolutional neural network [53]) on a large UI dataset (*e.g.*, screens, screenshots, web pages), to identify the features that are correlated to aesthetics to create a model to predict aesthetics [105]. Although this approach provides excellent results, it usually focuses on a specific goal, such as predicting the aesthetics of a specific category of user interfaces, such as user interfaces for electronic commerce. Therefore, the underlying model could be considered explicit and objective as it has been effectively created, but this model remains a black box that is challenging to interpret and impossible to modify unless completely retrained.

To overcome the limitations of methods for measuring UI aesthetics that are *implicit* (the model is embedded in the code or in the procedure), *invisible*, *not calculable*, and *inflexible*, therefore hard to interpret, this paper presents a quality model-based approach for measuring the ISO 25010 "User Interface Aesthetics" sub-characteristic by providing the following contributions (Fig. 1): a quality model objectively defines a quality function that accepts as input a set of UIs and that computes as output a value or a vector of values computed in terms of measures, units, formula, weights, thus making it *explicit* and *modifiable* by editing the configuration. The approach is *calculable* through GRACE, a software that enacts the quality model according to the quality function defined, thus making it computable and reproducible.

To this end, the remainder of this paper is structured as follows. Section 2 provides a state-of-the-art related to UI aesthetics, with a focus on the difference between qualitative and quantitative assessments and their impact on the perceived and actual usability of a UI. Section 3 provides a mathematical definition of the quality model and its quality function inspired by a definition scheme classically found in machine learning and illustrates some model variations. Section 4 presents the process of evaluating the aesthetics of the UI with the model-based quality approach and describes GRACE, a collection of web services that supports the execution of this process by computing the quality function. GRACE already implements 17 measures resulting from a systematic literature review (Appendix A.1) and can be extended to incorporate other measures when needed.

Section 5 reports on the results of a large-scale study conducted on a previous set of experiments whose purpose was to collect aesthetic scores from websites from end users. This study was carried out to assess the relevance and representativeness of each aesthetic measure and to illustrate the usefulness of the quality model-based approach. Section 6 discusses the benefits and limitations of GRACE. Section 7 concludes the paper and suggests some considerations for future work.

2 AESTHETICS AND USABILITY

Why should we measure the aesthetics of the user interface? Aesthetics are designed to attract users' eyes to the user interface and to make it easy and efficient interactional. While the contents are of utmost importance, the form is equally essential. Design appreciation is the initial interaction people have with a UI when they first face it: this period of time takes less than half a second

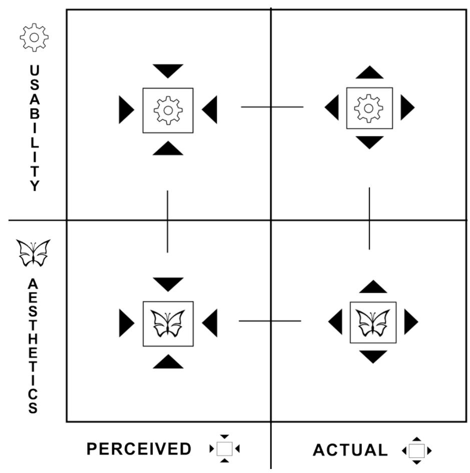

Fig. 2. Relations between the characteristic "usability" and its "user interface aesthetics" sub-characteristic.

[57, 76]. Furthermore, perceived UI quality has a positive impact on the idea users have about the usefulness and usability of the system [104]. Therefore, aesthetics is a potential element to optimize for improving the interaction with the end user.

Nowadays aesthetics has become an essential UI quality property, no matter the purpose. Whether they are designed for social networks, information websites, mobile applications, desktop applications, or games, they remain the first interaction modality provided to the final user to actually use the system. To reveal this importance, this section summarizes the relationship between aesthetics and usability, the former being a sub-factor of the latter [41]. The aesthetics of the interface is often strongly related to usability due to the effect produced by the aesthetics of a design on its perceived usability [9, 33, 74, 76, 91, 93, 94]. To keep usability in the loop, we use a classification key that represents the aspect of UI that the authors represent on the basis of two main properties ("usability" = and "user interface aesthetics" =), which themselves are related to two possible values (*extrinsic* = or *intrinsic* = – *i.e.*, perceived or actual). Therefore, we have four possible property values that are perceived usability = , actual usability = , perceived aesthetics = , and actual aesthetics = . Different relations can be established between these keys. For example, a reference discussing the effect of perceived aesthetics on perceived usability or another discussing the degree of difference between perceived usability and actual usability. The aesthetics of the user interface is considered a key factor that has a direct relation to (Fig. 2):

- The *perceived usability* of the user interface (). Ben-Bassat et al. [9] presented questionnaires in an experiment evaluating low and highly aesthetic UIs from an aesthetic and a usability perspective and found that perceived usability and perceived aesthetics were interdependent. Although performance was somewhat lower with the aesthetic UI, the respondents still perceived it as more usable than the non-aesthetic one. This suggests that aesthetic design makes things appear more usable, an idea argued by Norman [74] and Tractinsky et al. [93] in their paper "what is beautiful is usable". Tuch et al. [94] suggest that the UI beauty does not affect perceived usability, but that the increase of usability would increase the beauty of an application. Hassenzahl [36], Tuch et al. [94] take an opposite stance with what was previously mentioned and instead transform it into a "What is usable is beautiful" statement. However, it does not mean that one is right and the other is wrong but mainly that aesthetics and usability are inextricably linked, making it difficult to dissociate. This phenomenon was encountered by Papachristos and Avouris [76] in their experiment where end users assessed aesthetics before usability, even when they were instructed to describe usability [51]. More recently, Schrepp et al. [86] suggest that visual clarity possibly explains the dependency between perceived aesthetics (since clarity affects UI aesthetics) and perceived usability (with a possible effect on efficiency and learnability).

- The *effective usability* and its testing ( ). Hamborg et al. [33] studied the effect of aesthetics on perceived usability and the effect of usability on perceived aesthetics in the context of smartphone design. They developed a prototype with four different versions to test their hypothesis: low *vs.* high usability, and low *vs.* high aesthetics. 88 participants were asked to complete four different tasks. The results have shown no effect of aesthetics on perceived usability while there was an impact of usability on perceived aesthetics.
- The *overall impression* ( ). To investigate the first impression of web pages, Schenkman and Jönsson [85] found four important dimensions that impact the first perceptions: beauty, illustrations *vs.* text, overview and structure. They suggest that web designers should strive to give to their page a good overall impression by structuring the content and by making use of more illustrations than text to preserve an acceptable level of complexity.
- The *subjective user satisfaction* (→). Tractinsky et al. [93] experimented with the relations between users' perceptions of the beauty of a UI and its usability. For this purpose, they developed nine ATM UIs and asked 132 students to characterize both aesthetics and usability of these prototypes. By comparing results before and after the users actually used the nine UIs, they concluded that some aesthetic UIs, which were in reality not so usable, were not only perceived as usable before, as expected but also after they were used. This finding suggests that people are continuously affected by aesthetics when judging the usability of a system, thus concluding that "what is beautiful is usable".
- The *visual search* performance ( ). Salimun et al. [84] computed six aesthetic measures, *i.e.*, cohesion, economy, regularity, sequence, symmetry, and unity, and analyzed their values based on forced-choice paired comparisons to conclude that some measures, such as symmetry and cohesion, are more influential than others. They evaluated the relation between aesthetics and the performance of a visual search, showing differences between response times at each aesthetic level: the higher the aesthetics, the less time it takes to complete the required tasks.
- The *first impression* ( ). In their seminal paper entitled "Attention web designers: You have 50 milliseconds to make a good first impression!", Lindgaard et al. [57] conducted three studies to ascertain how quickly people may form an opinion about web page appeal. In the first and the second study, they presented web pages to participants for a time of 500 msec each. The results of both studies were close as the second was mostly a replicate of the first one. In the third study, web pages were no longer presented for 500 msec, but rather 50 msec. They obtained results highly correlated with the results from the first and second studies, thus concluding that visual appeal can be assessed by users within 50 msec.
- The *perceived performance* (→) These arguments can also be used to another universe than only one involving design and packaging since it had already been demonstrated with the "halo effect" [25] that people considered physically attractive had more chances to be perceived as competent and successful. Higher aesthetic treatments produce higher judgments of credibility in users' minds [82].

3 DEFINITION OF A QUALITY MODEL AND A QUALITY FUNCTION

Inspired by the classical definition scheme found in machine learning [54], we define a *quality model* as a mapping function expressed as:

$$Q_M : \mathbb{R}^{m\times n} \to \mathbb{R}^{\mathbb{R}^l} = X \to Q \tag{1}$$

where Q_M accepts as input a matrix of m features (lines of the matrix) and n values (columns of the matrix) of these features and returns as output a *quality function* $Q \in \mathbb{R}^{\mathbb{R}^l}$ defined as follows:

$$Q : \mathbb{R}^l \to \mathbb{R}^p = x \to q \tag{2}$$

The quality function Q accepts as input a vector of l *measures* and returns as output a vector of results of p values. If the quality function aggregates all measures into a single final value, then p=1, the vector becomes a value representing the final score, such as in SUS [17]. However, certain quality functions may need to return more than one single value when they consider that reducing a whole assessment process to a single value is oversimplifying. By definition, this formulation of quality models is formal and generic enough to encompass a wide variety of existing models. Calculability is a property that quality models defined by the formula 1 can have, but it is not always guaranteed. We can find quality models based on 1 that are not calculable, such as:

$$Q_M : \mathbb{R}^{m\times n} \to \mathbb{R}^{\mathbb{R}^l} = X \to \begin{cases} Q_T & \text{if } Q_M \text{ halts on input X} \\ Q_F & \text{otherwise} \end{cases} \tag{3}$$

where Q_T and Q_F are two arbitrary quality functions following the definition given by formula 2. Explicitness is not a property of the quality model as such but is rather linked to the whole process of model construction, including data collection and interpretation. The following subsections provide several examples of quality models to illustrate the versatility of the proposed formulation.

3.1 Application to SUS Evaluation Method

Questionnaire-based methods, such as SUS [17], can easily be expressed in terms of equations 1 and 2. For example, SUS methodology for a single questionnaire can be expressed as follows:

$$Q_{M-SUS} : \varnothing \to \mathbb{R}^{[0,7]^{10}} = \varnothing \to Q_{SUS} \tag{4}$$

where Q_{SUS} is the quality function defined as follows :

$$Q_{SUS} : [0,7]^{10} \to \mathbb{R} = x \to q \tag{5}$$

The quality model Q_{M-SUS} does not take any argument, since the quality function $Q_M - SUS$ is solely based on the questionnaire, and hence the domain of the function is defined as the empty set. The quality function Q_{SUS} is defined in the domain $\mathbb{N}^{10}$, corresponding to the feature space formed by the responses to the 10 questions that make up the SUS questionnaire. Here, we can assume a numerical encoding of the responses without loss of generality. The output of the quality function is a real number q corresponding to the score given by the SUS methodology. The SUS methodology expressed in terms of equation 4 is formal and also calculable as clear rules for computing the overall score are given. Explicitness, however, is not guaranteed as this property depends on the documentation of the processes underlying the methodology. The Explicitness of the SUS methodology based on a single questionnaire as in this example is trivial.

3.2 Application to SQuaRE-based models

Quality models based on the approach proposed in the ISO 25010 standard [41] can also be modeled in the formalism of equations 1 and 2. To illustrate this approach, we will consider a simplified version of the quality model based on three factors. The SQuaRE-based model can be expressed as follows:

$$Q_{M-SQuaRE} : \varnothing \rightarrow \mathbb{R}^{\mathbb{R}^7} = \varnothing \rightarrow Q_{SQuaRE} \quad (6)$$

where Q_{SQuaRE} is the quality function defined as follows:

$$Q_{SQuaRE} : \mathbb{R}^7 \rightarrow \mathbb{R} = (x_1, ..., x_7) \rightarrow q = \Sigma_{i=1}^{2} x_i \times \Sigma_{j=3}^{5} x_j \times \Sigma_{k=6}^{7} x_k \quad (7)$$

The SQuaRE-based model expressed in terms of equation 6 is formal by definition. It is also calculable, as all the operands and operators of the syntactic tree are defined. However, in the most general case, we should ensure that all the operators implied in the model are calculable in order to achieve the calculability of the model. Explicitness, again, is not guaranteed, as this property depends on the documentation of the processes underlying the methodology. Explicitness of the model is again straightforward, as all the parameters and inputs of the model are defined, but more complex models could imply the explicit documentation of parameters and/or inputs of the model.

4 EVALUATION PROCESS AND SOFTWARE SUPPORT WITH GRACE

Based on the previous quality model, the process for evaluating such a model in general and a specific one for UI aesthetics consists of two parts: the data collection (also called *measurement*) and the transformation of measurement values into measures (also called *calculation* or *metrication*).

4.1 Automatic and Semi-Automatic Process

While the calculation part is naturally automatic, the measurement part could be performed semi-automatically or fully automatically, resulting in a subjective or objective measure [67]:

- **Semi-automatic method**. A third party, such as a designer, an experimenter, is involved in the measurement process, not in the calculation, thus implying some subjectivity in the results. One person could measure the aesthetics of a particular UI and her perception of what has to be measured might be different from the perception of another person who is asked to accomplish the same task. There is even no need for the persons to be different as several judgments can arise from different contexts, *e.g.*, if I am in a hurry I will be more prone to measure things faster leaving behind some important elements. The same task, executed by different subjects and/or in various contexts, is likely to produce different results.
- **Fully Automatic Methods**. There is no intervention of any third party to tailor, to parametrize, or to focus the evaluation on some aspects. The objectivity of evaluation is then ensured since the system alone is collecting the data in a measurement phase and processing the measures into a computation phase. Ideally, efforts should be put into this type of method as it consistently produces the same responses regardless of any external influence.

Our method to analyze UI aesthetics is composed of four stages.

(1) *Select interface*: choose a UI in a defined context of use [19] (*e.g.*, for a user, a device/platform with some screen resolution, rendering the application, in a given environment).
(2) *Detect UI regions of interest*: detect and draw regions of interest on top of the previously selected GUI in order to retrieve basic forms attributes (*e.g.*,, position, dimensions) from the geometric plane.

(3) *Analyze UI aesthetics*: compute the UI aesthetic measures with regions attributes, interpret the quality model in which these attributes and measures are employed, and discuss and interpret results obtained in a measured report.
(4) *Redesign UI*: proceed to design modifications considering the previously established report.

This process can be repeated until the designed UI reach an admissible level of satisfaction. Step 1 is always manual. Step 2 could be achieved manually based on the designer's intervention or automatically by software that automatically identifies these regions based on some algorithm. The automatic step guarantees some consistency, while the manual approach is reserved for use by a designer who is experienced enough to determine regions on demand, thus achieving some more flexibility. Chen et al. [21] compared classical edge detection methods to deep learning methods for this purpose to conclude that deep learning outperforms past techniques.. Step 3 could be achieved manually or automatically depending on the agent that will compute the measures. Supporting this step automatically is, of course, a requirement. Similarly, step 4 could be achieved either manually (the designer herself moves, resizes, and modifies the GUI layout) or automatically (the system optimizes the layout). Automatic re-layout is outside the scope of this paper and will therefore not be handled [47, 60, 69]. We now summarize the main computations performed by Grace throughout this process.

4.1.1 Saliency Based Measures Computation. The saliency of an item is the state or quality by which it stands out relative to its neighbors. Saliency detection algorithms attempt to reproduce the human perception of image elements and to assess their visibility. Therefore, saliency is a measure of how a specific object is highlighted in comparison with the other objects presented to him. Long story short, it measures what part of an image is the most likely to be noticed. Several algorithms have been created to calculate the saliency of an image [18, 34, 38, 42, 49, 50, 104].

Shen and Zhao [88] were first to approach a visual attention model on web pages based on saliency. They use a Multiple Kernel Learning (MKL) model to predict eye fixations on web pages. In order to compute UI balance, we chose to apply saliency detection to first render a UI simplification to the elements composing its most remarkable parts.

4.1.2 Edge Based Measures Computation. Edge detection, or border detection, aims at identifying points in a digital image at which the image brightness changes sharply or, more formally, has discontinuities. Basically, and as the name indicates, it detects the borders of objects. A threshold can be used to apply various accuracy levels. According to Ziou and Tabbone [106], edges in an image correspond to discontinuities in the visual properties of scene objects and carry important visual information. Simply put, edges are significant local changes in an image [46]. Most of these edges are characterized as steps defined as inflection points occurring in an image.

Other types of edges are either line –a local minima– or *junction* - a meeting point of two or more edges. Multiple methods exist for border detection and are often varying in terms of filters, also called operators [62]. The geometry of the chosen operator determines to what type of edges direction it is most sensitive *e.g.*, vertical, horizontal, diagonal. Edges detection methods are generally categorized in two categories:

- *Gradient-based or search-based methods*, where edges are detected by looking for local maxima and minima in a first-order derivative expression of an image - the gradient direction.
- *Laplacian-based or zero-crossing-based methods*, where edges are detected by looking for zero-crossing in the second derivative of an image.

Regarding edge detection methods, the principal difficulty encountered by algorithms is dealing with noisy images, *i.e.*, images where borders are not clearly distinctive due to random variations of brightness or color information induced by the low quality of capture. In order to reduce this effect, the most common strategy is to pre-process the image through a smoothing phase - using, for example, Gaussian blur as it is done by Maini and Aggarwal [62] who produced a comparison of various edge detectors where he compared classical search-based operators such as Sobel and Feldman [90] or Roberts [81] with zero-crossing operators such as Canny [20]. The results showed that under noisy conditions, the Canny-edge detector exhibited the best performance.

4.1.3 Color Extraction Methods. Color extraction is relatively simple per se since it would require an algorithm capable of detecting all the different colors assigned to each pixel and storing them into an array of colors. The higher value for one specific color in this array would then stand for the color that is the most represented in a specific image. If this method could be applied easily in an algorithm in order to get a result that would be good enough, performance would be poor on common photos and images presenting a color depth of 24-bit RGB allowing to select of one color into a set of 16,777,216 possibilities (256^3). Therefore, with the necessity to find a solution to reduce this set of colors came out the process of color quantization, which simplified goal is to reduce the number of distinct colors in an image by selecting only restricted colors that are the most similar to the original set and to map the rest of the colors to them; hence enabling to keep a new image visually similar to the original one. Heckbert [37] invented a median cut algorithm that basically creates clusters of colors that are the most similar. The median cut process is as follows:

(1) Find the smallest box which contains all the colors in the image.
(2) Sort the enclosed colors along the longest axis of the box.
(3) Split the box into 2 regions at the median of the sorted list.
(4) Repeat the above process until the original color space has been divided into 256 regions.

As the resulting palette of colors is generally determined in advance, the method to find the nearest color is to minimize the Euclidean distance between each color of the original set and those of the resulting palette, resulting in a Voronoï diagram [99].

Fig. 3. Visual Complexity Interpolation.

4.1.4 Compression Based Measures Computation. Image compression is used to reduce the redundancy of an image in order to reduce its storage size. This is a commonly used technique to fasten images loading on web pages while keeping a level of quality that is good enough. The idea is to merge information that is perceived as similar on an image as unique and therefore avoid storing it several times, which would be inefficient. Simply put, the goal of image compression is to use a compression rate in order to render an image with the best quality possible while having the smallest size possible. As compression generally involves color quantization and other methods for scanning and removing pixel patterns that are similar and repeated in an image, it might be interesting to use its results as a predictor of visual complexity. Compressing an image enables us to consider some parts of the image that are easier to compress - *i.e.*, *simple* - and some parts that are difficult to contribute to a minimization of the image size - *i.e.*, *complex*. Tuch et al. [94]

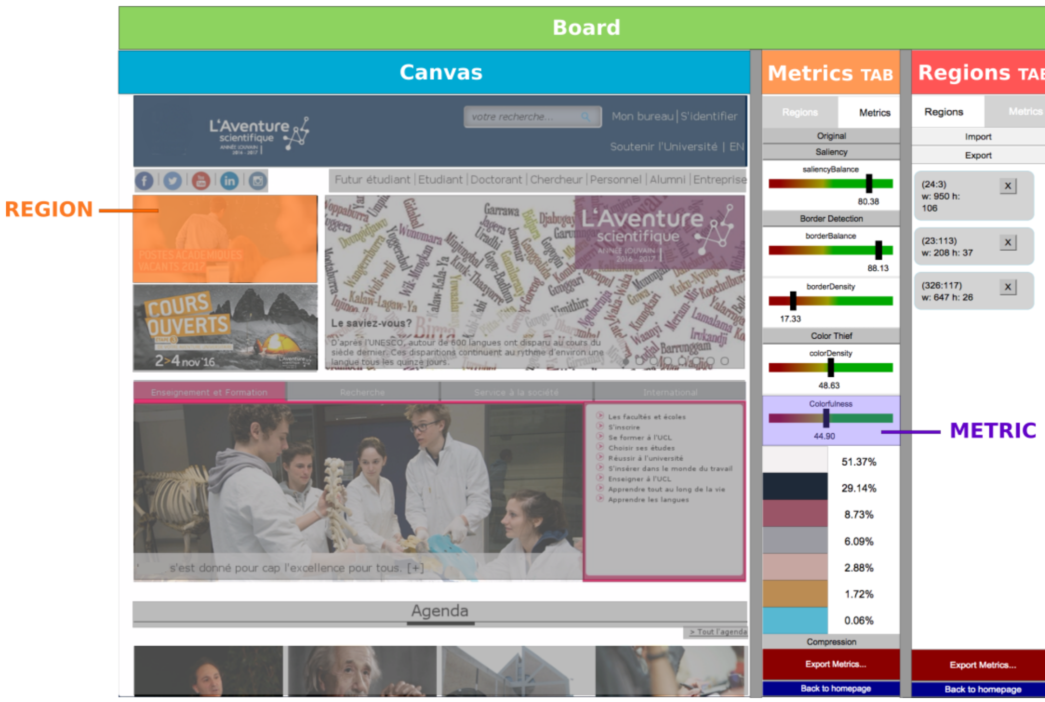

Fig. 4. Structure of GRaphical AesthetiC evaluation (Grace).

proposes more concretely to use the file size obtained after the compression of an image in order to assess its complexity. Intuitively, a complex image could be defined as an image containing lots of small elements varying in size and colors making them relatively different from each other and therefore harder to compress. The size of such an image even after compression should be high. By contrast, an image where elements are few and relatively similar in color while using the same time small palette would be much easier to compress resulting in a smaller file size.

Riglis [80] suggested measuring complexity through a ratio of bytes that are preserved after compression. Donderi and McFadden [26] conducted experiments that used compressed file length to predict search time and errors on visual displays used for marine electronic charts and radar. They found out that lossless JPEG file size was a good predictor of search performance and subjective complexity judgments. Another way of judging complexity, which is slightly different from what was above mentioned, is to compare an image to both an image that would be judged as completely simple - a blank image - and to another that is considered fully complex - an image including random colors for any pixels, making it almost impossible to compress. Interpolating an image size between these two images as depicted in Fig. 3 could help to produce a complexity measure.

4.2 Implementation of Grace

The software has been developed with the aim of implementing and validating the suite of measures (see the appendix for pseudocodes). Grace (*GRaphical AesthetiC Evaluation*) is an online web service evaluator that assesses user interface aesthetics based on measures. Grace is a web service generated with the Google Web Toolkit (GWT). The source code is written in Java and is compiled to JavaScript. The tool's implementation is currently separated into two main parts: the *client* where the user will be able to load a UI screenshot, define regions of interest, and consult a measures report, and the *server* where the majority of the computation is done by receiving region measures as an input and returning a report of measures as an output after the achievement of a metrication process. The *client* captures the user input and renders regions on top of a Canvas displaying the evaluated UI as an image as well as the measures report presented to the user. It is implemented in HTML5 and Javascript and makes it usable in any device with a browser. The *server* is implemented as a service that essentially provides the measurement computation. It also includes a RESTful API (Representational state transfer) allowing the server to be requested from an external source using a standard set of operations contained in a JSON input. The quality function is represented as a configuration file that is manually created as a structured JSON file structured as a tree of nodes, each node being a measure name along with a potential weight and operator such as addition, and multiplication. Editing the configuration file enables exploring another quality model and function.

5 APPLICATION OF A LARGE-SCALE STUDY

We conducted a last large-scale study based on results provided by Reinecke et al. [79] in a set of two experiments, the first which was to find a predictive model for the perceived visual complexity and color of websites and the second experiment conducted in the same set of websites with the purpose of collecting aesthetic scores from participants.

5.1 Lab in the Wild

Lab in the Wild is an online experimental platform developed to create surveys and collect participants' responses. The participants are not financially compensated but receive immediate information about how they performed compared to others.

5.2 Predicting Perceived Complexity and Colorfulness

In an online study, Reinecke et al. [79] collected visual complexity, colorfulness, and visual appeal perceptions of a set of 450 websites from 548 volunteers. The experiments were implemented as 10-minute online tests on the experimental platform Lab in the Wild. The procedure was to ask participants to rate 30 screenshots shown for 500 ms each. The small exposure time enabled to capture first reactions to the UIs complexity and colorfulness. Participants rated each website on a 9-point Likert scale from "not at all complex" to "very complex" or "not at all colorful" to "very colorful". In order to increase internal validity, a second phase required the participants after a short break to reiterate the rating for the same screenshots ordered differently.

Based on the data collected and on some automatically quantified web page properties such as colors, quad-tree decomposition and space-based decomposition representing sub-parts of the website, they computed various models for measuring visual complexity [105] and colorfulness [35, 102]. The regression model for the perception of colorfulness includes numeric values for various colors, saturation, a colorfulness value computed according to the method of Hasler and Suesstrunk [35], the number of image areas and quadtree leaves, as well as the number of text and non-text area. The regression model is highly predictive of the mean ratings of the colorfulness of the participants (r=0.88, p<0.001***, R^2=0.78, $R^2_{adj.}$=0.77).

The regression model for the perception of visual complexity includes numeric values for the text and non-text area, the number of leaves, the text groups and the image areas, a colorfulness value computed according to the method of Yendrikhovskij et al. [102], and the hue value of the screenshot. The regression model is highly predictive of the mean complexity ratings of the participants (r=0.80, p<0.001, R^2=0.65, $R^2_{adj.}$=0.64).

In the second experiment, researchers focused their survey on collecting website visual appeal preferences according to users. The same method was used to collect information, i.e., exposing 30 screenshots for 500 ms each and rating the website on a 9-point Likert scale. The resulting data amounted to 3412 ratings. The goal of this last experiment was to elaborate a model of website visual appeal based on complexity and colorfulness. Therefore, Reinecke et al. [79] confronted the complexity and colorfulness scores obtained through the responses of the participants to the first experiment with the aesthetics scores obtained in the second experiment. They found out that visual appeal was clearly influenced negatively by complexity and to a lesser extent by colorfulness. The bottom left of Fig. 1 shows some websites stimuli used in the experiment highly, medially, and lowly rated on their visual appeal score.

The impact of complexity and colorfulness on aesthetics is actually more complex than what can be extracted from a simple linear regression model, as it was previously suggested by Berlyne [10] and Tuch et al. [94] that the relation between complexity and a visual appeal followed an inverted U shape. Therefore, the quadratic terms of complexity and colorfulness were included in the model. The new results obtained indicate that this relation is not only negative - *i.e.*, the more complex the stimuli, the less appealing it is - but that it is also following a U-shape as expected both for complexity and colorfulness. The model has a R^2=0.48 stating that the aesthetic score is explained by complexity and colorfulness in a relatively important proportion.

Finally, a significant effect of age was detected to influence aesthetic perceptions of stimuli. Specifically, an interaction with visual complexity was discovered as participants older than 45 tended to find more pleasant websites with a low complexity level than other age groups. Another demographic that influenced the perception of colorfulness this time was the level of education. Participants with high school education were most negatively affected by high colorfulness and preferred websites with similarly low color. In a larger experiment involving participants from all over the world, Reinecke and Gajos [78] established that some components of visual appeal were strongly influenced by age, gender, geography, and education. Their conclusions are:

- The relation between visual complexity and visual appeal is represented by a low U-shaped curve. In other words, stimuli with high visual complexity results are less attractive, but stimuli with low levels of complexity are similarly liked to those with medium complexity.
- Visual complexity is the most salient website feature for visual appeal at first sight while perceived colorfulness only plays a minor role in people's first impression of appeal. This is explained by the fact that visual complexity somehow partly includes the influence of colors.
- Demographics may have an influence on aesthetic perceptions. In particular, education level significantly impacted colorfulness perceptions and age affected complexity perceptions. Gender can also be a discriminative criterion for aesthetic preferences in websites, as females liked colorful websites more and colorless websites less than males. Moreover, women also tend to dislike simple websites more than men.

5.3 Aesthetic Measurement with Grace

As a set of website stimuli had already been evaluated by participants in the previous study, we wanted to extend the available dataset with aesthetic measures computed with Grace. Therefore, we applied some implemented measures on the set of websites in order to assess if these measures might be good predictors of aesthetic preferences. In this study, the quality model is:

$$Q_{M-LabInTheWild} : \varnothing \rightarrow \mathbb{R}^{17^{\mathbb{R}^l}} = \varnothing \rightarrow Q_{LITW} \tag{8}$$

where Q_{LITW} is the quality function defined as follows :

$$Q_{LITW} : \mathbb{R}^l \rightarrow [0, ..., 1]^{17} = \text{screenshot} \rightarrow m = (m_1, m_2, ..., m_{17}) \tag{9}$$

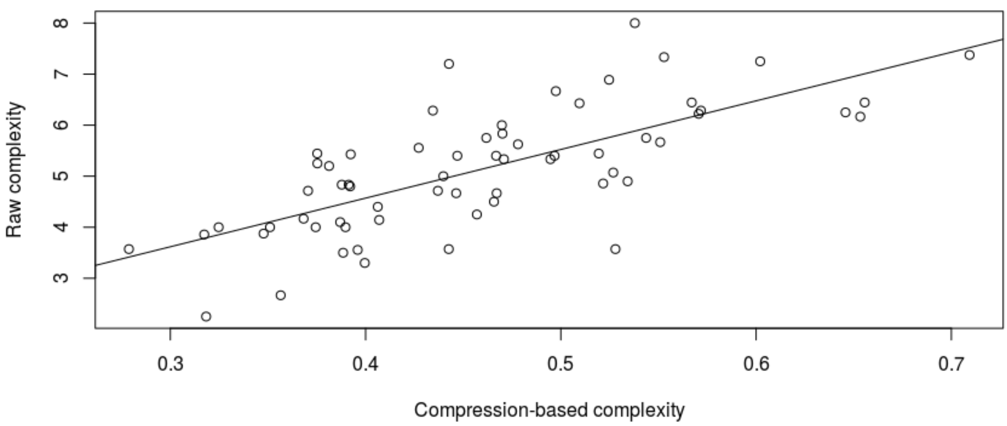

Fig. 5. Visual complexity of model fitting based on compression-based complexity.

where the domain $\mathbb{R}^l$ is the vector representing the space of screenshots and the co-domain $[0, ..., 1]^{17}$ represents the vector of all the 17 measures $m_i, \forall i \in 1, ..., 17$ computed for a screenshot, each returning a value normalized between 0 and 1. In this quality function, each measure is considered as having equal importance. A weight can be added to express a weighted average among measures.

5.3.1 Procedure. The first step was to evaluate each website screenshot through GRACE. In the first step, aesthetic measures such as colorfulness, density, saliency, and complexity were automatically computed. In a second step, after a manual definition of interest regions – a task executed by the evaluator – other layout arrangement measures such as concentricity, symmetry, and proportion were semi-automatically computed. Some measures were actually both automatically and semi-automatically computed such as balance and complexity with similar interpretations, only their methods and formulas were different. This gave a possibility to compare which method would give the best results. We proceeded with the region's definition of 100 website stimuli from the corpus of Reinecke et al. [79][1] (bottom on Fig. 1) and obtained 17 measures values for each screenshot. These measures were all those implemented in GRACE at the time of the experiment, namely global balance, vertical balance, horizontal balance, equilibrium, density, central alignment, external alignment, concentricity, simplicity, proportion, symmetry, saliency-based balance, border-based balance, border-based density, color-based density, colorfulness [35] and JPEG complexity.

5.3.2 Results. After having collected the measures for the aforementioned website stimuli, we were able to basically add them to the data provided by Reinecke et al. [79] that contained already 46 other measures. The data frame obtained merging our results then amounted to 63 collected measures. Of course, we were able to confront each measure with one another, but the interesting main point was to be able to confront our measures with the three main participants' responses which are the scores for visual appeal, complexity, and colorfulness. The most interesting result is undoubtedly the comparison between our compression-based complexity measure and the raw complexity values. Indeed, defining a model taking into account this only one variable for predicting

[1]See https://homes.cs.washington.edu/~reinecke/Publications_files/data/SupplementaryMaterials.zip and https://homes.cs.washington.edu/~reinecke/Publications_files/data/website_stimuli.zip

complexity gave a R^2 equal to 51%, as depicted in Fig. 5 - and since the correlation between both variables is 71%, we can conclude that our method for evaluating visual complexity of websites based on the JPEG compression size is not only simple but also appropriate. Note that the values predicted by our measure and the predictions of Reinecke's model are also highly correlated (r=0.72).

Adding several other automatically and semi-automatically computed measures presented in Table 1 from Grace, we were able to make the model even more predictive (R^2=0.69). Note that colorfulness, horizontal balance, and simplicity contribute negatively to a model of visual complexity prediction. It is no surprise that a measure of simplicity negatively impacts complexity as they are opposite by definite. The same type of conclusion can be drawn regarding colorfulness as grayscale stimuli are considered simple. A layout with objects equally balanced on its left and right parts is considered more complex. Probably, websites with lower complexity values tend to regroup their small number of objects in one horizontal part of the layout. Moreover, layout presenting a high number of objects tends to be more complex and paradoxically more balanced as an important number of objects implies subsequently another space distribution. Finally, center alignment contributes positively to complexity perception. Indeed, a layout composed of objects with various dimensions presenting a high degree of center alignment directly involves poor external alignment, which is likely to increase visual complexity.

Table 1. Complexity model : (semi-)automatically computed measures.

	Estimate	**Std. Error**	**t value**	$Pr(>\|t\|)$
(Intercept)	3.7138	0.8977	4.14	0.0001
Compression Complexity	8.3425	1.0249	8.14	0.0000
Colorfulness	-1.1054	0.4177	-2.65	0.0105
Center Alignment	2.6214	0.7251	3.62	0.0006
Horizontal Balance	-2.3166	0.6386	-3.63	0.0006
Simplicity	-2.8350	1.3655	-2.08	0.0424

Another possibility was to combine the best of both worlds regarding visual complexity prediction and therefore use the Reinecke predictive model of complexity and our compression-based measure of complexity. This resulted in a model that was sufficiently predictive (R^2 = 0.55), while presenting the advantage of being strictly limited to automatically computed measures.

Table 2. Complexity model: mix of complexity values.

	Estimate	**Std. Error**	**t value**	$Pr(>\|t\|)$
(Intercept)	0.2922	0.5759	0.51	0.6137
Compression Complexity	6.7739	1.6629	4.07	0.0001
Reinecke Complexity	0.3366	0.1463	2.30	0.0249

Finally, we investigated various models to find out which was the best fit for predicting aesthetic scores. Our resulting model is a Log-linear model as presented in Table 3 with a R^2 equal to 60%. This model includes Grace measures of compression-based complexity, balance, and concentricity, as well as measures representing number of Leaves and Quadtree Leaves computed by Reinecke et al. [79]. Other observations can be made by analyzing the correlations and results of the measures.

Table 3. Model of aesthetics based on a mix of measures.

	Estimate	Std. Error	t value	$Pr(>\|t\|)$
(Intercept)	2.0588	0.2428	8.48	0.0000
Compression Complexity	-1.3413	0.2736	-4.90	0.0000
Balance	0.6242	0.1912	3.26	0.0019
Concentricity	-1.0737	0.3397	-3.16	0.0025
Number of Leaves	-0.0015	0.0006	-2.47	0.0166
# of QuadTree Leaves (color)	0.0001	0.0000	2.98	0.0042

- Surprisingly enough, our compression-based complexity measure is also highly correlated with the raw colors (r=0.72). Actually, it is because the raw values of complexity and colorfulness themselves are highly correlated (r=0.89).This indicates that the perception of visual complexity is primarily concerned with color perception.
- White and silver color values are highly negatively correlated with the measure of color density. It is not surprising as the color density measure is based on the proportion between non-background colors and background colors. This clearly shows that the most used background colors on websites remain white and silver.
- Text area measure is fairly well correlated to the measure of compression complexity (r=0.50). This indicates the proportion of text area in a website amount to a good part of its complexity. On the other hand, the non-text area measure is quite correlated with colorfulness models (r=0.54) indicating that these non-text areas are generally colorful pictures. It is even more visible regarding the number of image areas (r=0.80).
- The semi-automatically computed measure of balance is best represented by the border-based balance measure (r=0.47) than by the saliency-based balance measure (r=0.13) which is itself much more an indicator of equilibrium (r=0.41). Semi-automatically computed density has a relatively low correlation level with border-based density (r=0.37) or with color-based density (r=0.27) which could be explained through the region area being drawn as a plain envelope in the definition of interest regions. This does not reflect exactly the amount of space being used by one region as it may contain blank space.
- Symmetry seems to have, to a lesser extent, a negative impact on visual complexity ($r=-0.36$) therefore, impacting aesthetics score positively (r=0.32).
- As the raw scores are generally collected among participants and as a majority of them tend to avoid responding with scores too far from the neutral response - that can also be characterized as a fear of the extremes - when they feel they are not experts, binomial regression might be considered in order to improve model predictions.

5.4 Comparison and Discussion

The experiments conducted by Reinecke et al. [79] produced significant results on a large set of stimuli with a large number of participants. These results were mainly intended to validate predictive models of website complexity and colorfulness and use these models as predictors of aesthetic judgment scores. That aim has surely been achieved as the researchers validated a model of complexity based on various measures such as the proportion of text and non-text areas and colors obtaining a high coefficient of determination. The same conclusion can be drawn for their measure of colorfulness perception based on a mix of numerical values of various colors and the results of a colorfulness algorithm proposed by Hasler and Suesstrunk [35]. Another important finding achieved through this study is that the aesthetic score is explained by complexity and colorfulness in a relatively important proportion. The relation existing between them in the quadratic model shows that it follows a slightly inverted U-curve. To extend the study, we computed another set of measures with GRACE in order to gain a new perspective and assess the predictor quality of the implemented measures. We processed 100 websites and obtained a set of 13 measures for each website (Table 4, thus enabling a comparison with the original data. Our model for complexity based on image compression size is highly correlated with the complexity judgments collected by Reinecke et al. [79] and a linear model using this only variable results in rather good predictions. Regarding the prediction of aesthetics score, we propose a log-linear model incorporating a mix of measures chosen in the global set gathering our proposals and measures used in the original study. This model of aesthetics is highly predictive and the measures that are the most impacting it are complexity (-), balance (+), and concentricity (-).

5.5 Validation Criteria

In terms of validation, we defined two important aspects to focus on when attempting to validate aesthetic measures: (1) the relevance as the extent to which a measure is an important aesthetic criterion - a measure considering, for instance, the number of spelling errors in the text content of a website might be of high importance for assessing the professionalism and literacy level of its authors but would not be relevant at all when it comes to evaluating first aesthetic impressions; and (2) the representativeness as the extent to which the value provided by a measure is representative of what a human being may assess - inventing a formula is an easy task as it is possible to gather random measures with some arbitrary coefficients and to produce this way a measure supposed to quantify a designated attribute but it is more difficult to know if that measure would really measure what it is supposed to, it is not by combining ingredients randomly that you know how to cook. Therefore, we can classify the experiment conducted as an attempt to tackle both the relevance and representativeness issues of the measurement-based evaluation. It may be observed that the representativeness issue is the first issue considered. Namely, by providing empirical validation for a measure through user reviews, we looked for strengthening its reliability.

On the other hand, the issue regarding the measure relevance has been solved as we have found empirical evidence that some measures were important factors to take into account in design in order to reach an acceptable level of aesthetics. These measures are complexity, balance, and concentricity based on the study with the most significant results. Table 5 indicates relevance and representativeness for each measure in the considered experiment. As the initial purpose of the large-scale study was mainly to collect human judgments about visual complexity and colorfulness, representativeness analysis was not applicable (NA) for other measures. Another important point that is worth mentioning is the reproducibility of the experiment in a given population. Results obtained from this large-scale study are significant with a number of 450 UI stimuli rated by 548 participants. In fact, 30 UIs were shown to each participant selected randomly in a set of 450.

5.6 High Priority Measures

The experiments discussed above enable us to consider some priority in aesthetic measures for aesthetics. Firstly, it is possible to determine which visual aspects of a user interface should be considered thoughtfully during design. Therefore, we retain first and foremost visual complexity. It is the measure that is the most representative of user reviews and that is the most impacting aesthetic consideration of a design. Indeed, we noticed, as was the case for Reinecke et al. [79], that the relation between complexity and aesthetics follows a slightly inverted u curve where very low and very high values of complexity are responsible for low values of aesthetic considerations among participants. The impact is even more salient for high values of complexity, while intermediate complexity tends to guarantee medium and high scores of aesthetics.

Table 4. Thirteen measure formulas used to compute scores for each interface.

Measure	Formula
Alignment	$A = \frac{DA_v + DA_h}{4n}$
Balance	$BM = 1 - \frac{\lvert 1 - \frac{w_L + w_R}{max(\lvert w_L\rvert, \lvert w_R\rvert)}\rvert + \lvert 1 - \frac{w_T + w_B}{max(\lvert w_T\rvert, \lvert w_B\rvert)}\rvert}{2}$
Euphemism	$\frac{\mu_{area}}{A}$
Symmetry	$1 - \frac{\lvert SYM_{vertical}\rvert + \lvert SYM_{horizontal}\rvert + \lvert SYM_{radial}\rvert}{3}$
Regularity	$\frac{\lvert RM_{alignment}\rvert + \lvert RM_{spacing}\rvert}{2}$
Simplicity	$\frac{3}{n_{vap} + n_{hap} + n}$
Economy	$\frac{n}{n_{size} + n_{colors} + n_{shapes} + n}$
Unity	$\frac{\lvert UM_{form}\rvert + \lvert UM_{space}\rvert}{2}$
Grouping	$\frac{\sum_{i=1}^{n}\sum_{j=1}^{n}\frac{S_{ij}(L+H)}{2D_{ij}}}{n}$
Density	$1 - \frac{\sum_{i}^{n} a_i}{a_{frame}}$
Predictability	$\frac{C + SQM}{2}$
Consistency	$\frac{3n}{n_{colors} + n_{shapes} + n_{proportions}}$
Continuity	$\frac{4}{\sigma_{dist} + \sigma_{shape} + \sigma_{size} + \sigma_{orientation}}$

Other visual measures should include balance and concentricity. Only the horizontal balance seems to be the primary focus of attention, which is in line with our human vision mostly influenced by the horizontality in the eye position. Regarding concentricity, a layout where the objects are closer to the center is preferred to a layout sticking to the borders. This aspect is generally tackled by the document margins which provide some space between the border and the content, inducing a sense of clarity. Conversely, elements too close to the layout center would create some unnecessary local density, and the effect would be just the opposite. It must be emphasized that these aspects are those related to measures classified as relevant according to the aforementioned experiments. It is clearly evident that there exist other important visual principles to apply when designing user interfaces. Principles such as alignment, spacing, density, and minimalism may also be considered.

6 DISCUSSION

6.1 Context

This paper addresses the implications of UI aesthetics as a potential element to optimize in order to capture the user's interest. Multiple research activities have shown that perceived quality of a UI has a positive impact on the idea users have about the system's usefulness and usability [9, 104]. This perceived quality is, of course, related to the satisfaction that the user can draw from the usage of a specific GUI, but only to a lower extent compared to the impact of the perceived aesthetics of the system [85] advocating for the famous motto that states that what is beautiful is usable [93]. The benefits of UI aesthetics should not be demonstrated anymore. The central question was then: "How to assess UI aesthetics?". The qualitative evaluation of a design provided by several user judgments remains an appropriate solution to improve UI design. However, this kind of evaluation process is cumbersome and can often only be undertaken when the UI is already released. A better approach for the first design evaluation could be to objectively quantify aesthetics through an evaluation by aesthetic measures. The interest in such an approach lies in the quick and objective feedback that the designer may receive. Indeed, rather than having to gather people and optionally conduct a survey to receive comments on some way to improve the UI aesthetics, she may simply define UI regions and request a measurement report. Afterward, she has the necessary elements to modify the GUI in order to improve its aesthetics. In the following section, a summary of what has been presented in the document to better address this matter is developed.

Table 5. Measures conclusions for the large-scale study.

Measure	Relevant	Representative
Center Alignment		NA
External Alignment		NA
Balance	X	NA
Vertical Balance		NA
Horizontal Balance	X	NA
Saliency Balance		NA
Border Balance		NA
Complexity	X	
Compression Complexity	X	X
Concentricity	X	NA
Colorfulness		X
Density		NA
Border Density		NA
Color Density		NA
Equilibrium		NA
Proportion		NA
Symmetry		NA

6.2 Limitations

Limitations may be made regarding the present research work. These limitations are classified with respect to the threatened validity of the research design. We explain the validity threats of our empirical study in terms of four categories: internal, external, construct, and conclusion threats to validity [101].

6.2.1 Threats to Internal Validity.

- No study was conducted exclusively with the purpose of collecting data among participants active in the field of product design. Indeed, interviewing professionals in the field would have represented an opportunity to better address the question of "what makes a UI design appealing?" and confront their vision with the measures explored in this work for validation. Therefore, a qualitative research design could be undertaken from this point of view in order to identify the key elements that distinguish beautiful and ugly designs.
- There is a need to provide more robust and significant empirical validation of how measures express aesthetic aspects through several experiments focusing on relevance and representativeness. As long as new aesthetic measures could be defined regarding various visual techniques, there will be space to investigate the representativeness of the measures. The relevance of the measures might also need to be supported with more empirical validation, as we can only currently conclude that balance, visual complexity, and concentricity of a design have an effective and significant impact on aesthetics. Though, these measures do not represent GUI aesthetic judgment in its entirety.

6.2.2 Threat to External Validity.

- The representativeness of the results depends on various parameters in the process, such as the size, the variety, and the complexity of the dataset, but also the diversity of the metrics might affect the generalization of the results. The results obtained for a particular quality model-based approach are certainly valid for the dataset considered and can be re-applied to another dataset. But these results can hardly become transposable to another context. For example, two quality models based on the same set of metrics, but with different formulas, are likely to give rise to two different, perhaps inconsistent, interpretations. This is where GRACE can play some role by testing and comparing different formulas on the same dataset. Beyond this comparison, results obtained for a dataset belonging to a single domain, say electronic commerce, are unlikely to be easily transposable to another domain, say online newspapers. The values of measures, their typical range, but also their distribution is very likely to change depending on the domain of activity.
- The representativeness of the results could also be affected by the number of GUI layouts analyzed. We believe that the GUI layouts selected for this study can be considered as a baseline to obtain some indications of which formulas return more representative results. In any case, the total number of GUIs considered in a dataset, whoever large it is, cannot represent the whole GUI design space. Replications with different and more complex GUIs are still needed to study the effect of their variables on the observed results, a process that is largely facilitated by the flexibility of the quality model-based approach. The big win is that, when the quality model changes, the results automatically change.

6.2.3 Threats to Construct Validity.

- The set of aesthetic properties being almost unlimited, it was difficult, except for those extensively supported in the literature such as balance and visual complexity, to determine what measures to retain and to implement with an order of priority. As it is possible to define measures like a cook can do it with a recipe, it is likely that we missed an important one that could be an effective indicator of GUIs aesthetics. This affirmation can even be supported by looking at the last model of aesthetics proposed based on complexity, balance, and concentricity in the large-scale experiment. The prediction of the aesthetic score model was highly representative of human aesthetic judgments with a determination coefficient of 60%, indicating that the origin of the remaining 40% has not yet been discovered.

- Improvements can be made in the way of addressing the model. At the moment, it only takes into account the arrangement of GUI elements in a layout. It is necessary [15, 71] to propose a model considering other potential GUI aesthetic aspects such as color combinations and shape complexity.
- When dealing with the validation of measures, only a summative evaluation has been considered: Since Grace displays the measurement values on top of a colored spectrum, not only the value (summative part) but also the positioning with respect to the acceptable range (formative part) could also be considered. Any UI element in Grace could be moved, resized, and realigned, allowing the designer to explore alternate designs based on the same elements or not. An optimization problem then arises in order to optimize the global layout under the constraints of measures being located in acceptable ranges. If solved, Grace would suggest what operation to perform on which UI element to reach an optimum or close by.
- The main limitation of the semi-automatic method is that it is not taking a sufficient distance from the subject and may lead to bias in the measures responses. Indeed, when a human is directly involved in the definition of the regions, the measures might somehow be impacted by the subjective perception of this definition as distinct persons would probably be prone to spot various interest regions for the same GUI; hence transforming the automatically computed measures into somehow subjective measures by their very nature.
- Another limitation is the lack of rules and constraints to apply in the definition of regions of interest. An item about that is dedicated in the next section dealing with future considerations.

6.2.4 Threats to Conclusion Validity.

- With regard to conclusion validity, the data collection and the validity of the statistical tests applied must be considered. With the purpose of decreasing the data collection threat, the same data-extraction procedure and the same data-calculation procedure were applied, which is guaranteed by Grace. The statistical tests are outside the scope of Grace. It is therefore our responsibility to select and apply the right tests for our hypotheses [65]. To select the statistical tests, we considered the design of the experiment and the nature of the variables and their assumptions according to Maxwell [64].
- A first threat is to conclude that there is no relationship when in fact there is some. This may be due again to the lack of random heterogeneity of GUIs in the dataset, but also of the participants. A very diverse group of participants is certainly desirable but could lead to diverse results which, ultimately, do not enable identifying a relationship.
- A second threat is to conclude that there is a relationship when in fact there is no such relationship. A not-so-diverse group of participants is less desirable but could lead to a more concentrated sampling of the population, which, ultimately, enables identifying a relationship. This relationship is however only valid for this small group and cannot be concluded for the population.

7 CONCLUSION AND FUTURE CONSIDERATIONS

To overcome the limitations of methods for measuring UI aesthetics that are *implicit* (the model is embedded in the code or in the procedure), *invisible*, *not calculable*, and *inflexible*, this paper presented a quality model-based approach for measuring the ISO 25010 "User Interface Aesthetics" sub-characteristic by providing the following contributions: a quality model is objectively defined in terms of aesthetic measures, units, and measurement methods (thus making it *explicit* and *modifiable* by editing), the evaluation method is *calculable* through GRACE, a software that computes the quality model according to the method, thus making it computable and reproducible.

With the purpose of developing methodologies and tools using measurement validation, we have chosen to focus our efforts on the definition of a method and a tool (GRACE) in order to give concrete examples of what can be achieved in the design process when particular attention is paid to the evaluation of the aesthetics of the GUI. Therefore, improvements can be made in the way of addressing the model. It is necessary to propose a model considering other potential GUI aesthetic aspects such as color combinations and shape complexity. We made a step forward in this direction by proposing automatic methods for collecting measures based on the level of compression and on the level of colorfulness. It might be adequate to also investigate the effect of the use of color schemes in design (*e.g.*, a measure representative of the monochromatic, complementary or triadic properties of the color schemes that are used). Another suggestion is to take into consideration the fact that UI aesthetics vary with the domain in which it belongs. Indeed, it may be nonsense for some designers to achieve a UI in which objects are completely balanced. Therefore, an idea would be to split each measure in such a way that it represents no more one positive and one negative value (Balance/Unbalance) but rather two distinct components.With this point of view, it is possible to analyze joint effects on UI aesthetics (*e.g.*, effect of an unbalanced and misaligned UI).

In the scope of this work, we chose to only keep simple measures that can be categorized as measures of classical aesthetics in the sense of Lavie and Tractinsky [52], therefore excluding aesthetic measures rendering a more expressive meaning such as abstractness or sharpness which are characterizing a more esoteric art, not accessible to a lay audience, and probably less focused on the layout arrangement of objects but rather more on their individual properties. Ideally, aesthetic measures should be defined for all kinds of artistic phenomena when possible. Regarding the evaluation by measures with a semi-automatic method, we made a small step in providing a tool for the drawing objects of an interface and the request of a quantitative result. It would be interesting to automate most of this evaluation process and propose a system of recognition of shapes and colors. The interest would be, on the one hand, to avoid manual drawing of objects and, in this way, eliminate any subjective bias that could arise from an individual's own perception of the existing interest regions, and, on the other hand, the evaluation could be applied to any interface regardless of the language which is used for defining the interface. If the choice is done to keep a semi-automatic method, it might be relevant to establish some rules and constraints documenting when it is appropriate to define a region of interest and how deeply a region contained in another region should be considered as a unique region of interest. An idea could be to use a threshold based on the Gestalt law of figure-ground to determine whether a region is part of the background or is really put forward in the layout. Some measures may be optimized to better reflect human eye perceptions. For example, it is necessary to provide an accurate and validated threshold for categorizing correctly values of density that are considered efficient or inefficient from a human perspective, therefore, enabling one to tackle the issue of measures representativeness. Moreover, the thresholds used in the analytics panel of GRACE were arbitrarily defined. It is therefore necessary in a research process to provide empirical validation for these thresholds in order to propose a reliable evaluation tool. In the same vein, it is necessary to provide further empirical validation for the previously defined measures with a final goal of solving the representativeness and relevance issues raised by such a quantitative approach of aesthetic evaluation.

An attempt to solve the representativeness issue could be to reproduce the aforementioned study at a larger scale (*i.e.*, exploring a large corpus of interfaces with a large sample of users with regard to a large set of aesthetic measures) in order to support formulas represented for each measure. In another experiment, it could be possible to tackle the relevance issue by investigating the methods and principles of professional designers when it comes to assessing GUI aesthetics. This experiment could be in the form of a qualitative research design on the visual techniques designers use in their work with the purpose of understanding what are the rules and good practices, with the objective of not missing them when designing GUIs and spotting the new trends that are essential to ensure their modernity. To put it in a nutshell, this research design would involve semi-structured data collection based on interviews with professional graphical, web, and mobile designers. Questions in the interview guide should be directed toward what makes a GUI appealing or unappealing and toward what is their feeling about some existing visual techniques that are often cited in scientific literature. Confronting the results of such a survey to a set of measures would allow making a distinction between those that are relevant for aesthetic validation and those that are irrelevant.

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A A SYSTEMATIC LITERATURE REVIEW OF AESTHETIC MEASURES

A.1 Definition of Aesthetic Measures

To define a set of aesthetic measures comparable to BaLOReS [31] and similar software [44, 46], this appendix defines and applies consistently a structured method for determining aesthetic measures, along with their own definition and formula, if any. After performing a systematic literature review [16] for each individual measure, we propose our own definition, one or many formulas, and one or many interpretations of the measure based on the review and on our own reflection. First, we apply the Goal-Question-Measure (GQM) approach [7] to assign each aesthetic measure to a high-level question. The initial goal is shared by all measures: *an evaluation of the user interface aesthetics taking the standing point of the end user.* Therefore, the catalog presented in Table 6 only includes the Question-Measure association. The next subsection explains the process of searching through the literature for each measure.

Table 6. A catalog of aesthetic measures according to the GQM method [7].

Question	**Measure**
Is the composition of the UI easy to understand?	Simplicity, Economy, Understatement
Are the colors pleasant?	Colorfulness, Color Harmony
Are the objects on the layout harmoniously arranged?	Balance, Symmetry, Equilibrium, Alignment, Regularity, Repartition
Are the objects on the layout consistent in shape?	Proportion, Horizontality, Angularity, Stability
Is the UI clear and sufficiently guiding the user?	Density, Neutrality, Singularity, Spacing, Concentricity, Grouping
Are the UI objects ordered on the layout?	Consistency, Sequentiality, Predictability

A.2 Search Process

To compile, analyze, and compare existing definitions of aesthetic measures and formulas, we perform, for each measure, a Systematic Literature Review (SLR) [16, 48] following the PRISMA method [56] and review guidelines [89]. The search process aims at identifying references that are relevant within the research scope in a well-defined and rigorous way that could be reproduced. For this purpose, a semantic query is first defined independently of the digital libraries where these references could be found. This semantic query is then transformed into a series of syntactical queries expressed in the respective query languages of digital libraries. These syntactical queries are then executed to produce the input of this search process. Although the term "metric" is predominantly used in Human-Computer Interaction, we also add the term "measure" [55], which is now a standard term in software engineering and software testing, even replacing the "metric".

(1) *Definition of semantic query*: the semantics (*i.e.*, the words) of interest are defined to specify the desired search outcome (*e.g.*, the user interface measures of balance and complexity).
(2) *Definition of syntactical query*: A typical syntactical query takes the form of a suite of words separated by logical operators, *e.g.*,

```
SQ = (user interface OR layout OR aesthetics) AND (measure OR metric) AND
                        (balance OR complexity)
```

This latter query would gather results regarding an aesthetic measure of balance and complexity in the UI taking into account the cases where the user interface is also named layout and the metric is alternatively considered by measure, in accordance with previous ontologies [55].

(3) *Identification of single-publisher sources*: to obtain an initial set of potentially relevant references, a list of scientific digital libraries needs to be identified. Since UI measures lie within the broad field of computer science and overlap several domains of computer science, the digital libraries of the major computer science publishing houses were selected to ensure a partition of references, thus minimizing duplicates (apart from joint publications): Association for Computing Machinery Digital Library, Elsevier ScienceDirect, Elsevier Ei Compendex (COMPuterized ENgineering inDEX), Institute of Electrical and Electronics Engineers (IEEE) Xplore, Springer SpringerLink, CiteSeerX, Cornell University Library arXiv, U.S. Government science and technology library Science.gov, and Wiley Online Library. At this point, the restricted search of the ACM Digital Library, which selects only references published by ACM Press and not affiliated organizations, such as IEEE, was used to maintain the intended partition.

(4) *Identification of multi-publisher source*: to ensure the quality of the references on UI aesthetics, to validate these search results and to cover other publishers, we complemented the five single-publisher sources with three generic search engines that cover multiple publishing houses and sources: ACM Digital Library Advanced Search (including all publications indexed by ACM as well as affiliated organizations), DBLP CompleteSearch and Google Scholar. They have been used more as a verification mechanism than as a pure search process.

After a first series of query execution, order of results according to relevance and removal of duplicates [56], rules of inclusion relative to content were applied [16, 89]. This included all references complying with at least one of the following conditions:

- (1a). The abstract or introduction explicitly mentions user interface aesthetic measures or an intended synonym as previously identified.
- (1b). The abstract or introduction explicitly uses the technical terminology used as an intended synonym for user interface aesthetics as previously identified. Condition (1b) ensures that no technical references, mostly implementation-oriented, are excluded within a particular technological space, but nevertheless address some calculation of an aesthetic metric. Additionally, the following mandatory conditions need to be fulfilled:
- (2) It can be deduced from the abstract or introduction that the research is performed within the context of software engineering and within the context of UI aesthetics.
- (3) It tackles one or several research topics relevant to UI aesthetics as the main topic.

To proceed to each SLR for each measure, we established a set of frequent words that would compose our queries made to the search features of digital libraries (Table 7). Sometimes words appearing in references resulting from one query were incorporated into the set to deepen the understanding of the related concepts. The results of these SLRs are presented in Appendix A. For

Table 7. Set of concepts related to aesthetic measures.

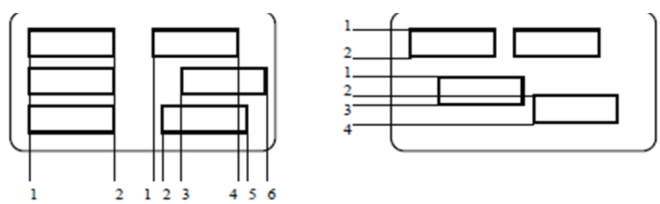
Measure	Aesthetic	User Interface	Measure	Evaluation
Quality	Design	Layout	Alignment	Visual
Complexity	Gridedness	Balance	Equilibrium	WeightMap
Saliency	Symmetry	Order	Centricity	Craftsmanship
Figurativity	Density	Grouping	Minamilism	Euphemism
Simplicity	Similarity	Colorfulness	Proportion	Orthogonality
Prototypicality	Contrast	Clarity	Color Schemes	Gestalt

each aesthetic measure, a general definition is provided, followed by a timeline of references that deal with the topic, and a detailed discussion of each reference.

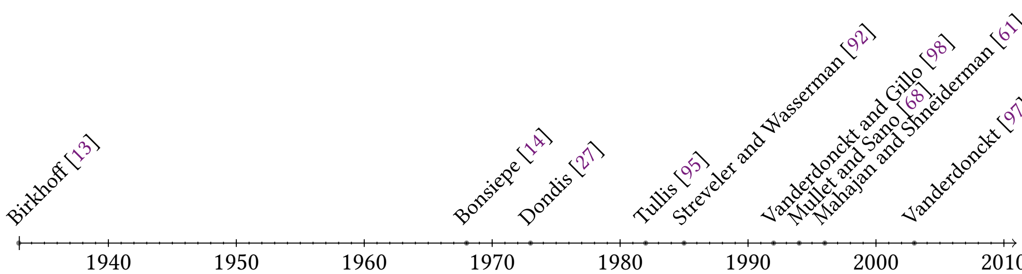
B RESULTS OF THE SYSTEMATIC LITERATURE REVIEW

B.1 Alignment

Alignment in UI is mainly the extent to which a design has its items placed in a way that they are lined up on their edges or on their centers. It is a visual technique commonly used to organize an interface, but more than that, alignment has been used in almost every kind of design since the ages.



In writing, letters are aligned and grouped in order to form words themselves aligned with other words to form sentences. Alignment in writing design is a technique that has been used for a long time.



A first measure of polygon alignment is proposed by Birkhoff [13] in his work on the aesthetic measure. This measure is not of direct importance as it is valid for one polygon at a time and not the arrangement of a set of polygons - which is what compounds most of the screen's designs - but it has the advantage to be computable and gives an idea on how alignment must be formalized considering alignment as a measure of how a polygon is aligned on a horizontal-vertical network. Tullis [95] is the first author to approach alignment in screen design. The main concepts approached are around a vertical alignment, since at this time interfaces were mainly composed of alphanumeric displays. A first related guideline recommends that words should be left-justified, while numeric data should be right-justified in order to increase predictability of item locations. Actually, there exist well fragments of alignment consideration in research prior to Tullis [14, 27].

However, those were more focused on layout complexity (including regularity, distance, and balance measures) than solely alignment measures. Streveler and Wasserman [92] are probably the first authors to propose a quantitative measure for columnar alignment. Two main reasons are invoked to support alignment. First, it implies list processing, which, according to the authors, defeats our impulse to read. Second, it provides a visual straightedge along which our eyes can accurately travel. According to Mullet and Sano [68], an aligned position of UI elements reduce the screen complexity, thus improving other aspects like the visual search time of an item on this screen, hence contributing to a cleaner and more understandable UI. Moreover, the authors argue that aligned objects, even if separated by large distances, seem to attract each other and, therefore, create a visual relationship between them.

AIDE, a software tool developed by Sears [87], implements measures to evaluate the UI structure, including alignment. The goal of this tool is to provide measure-based assistance in the UI creation process. *Gridedness* is the term used by Mahajan and Shneiderman [61] to describe a measure to align UI widgets. The authors distinguish two components forming the alignment: *X-Gridedness*, that counts the number of sets of widgets with the same X coordinates and *Y-Gridedness*, that counts the number of sets of widgets with the same Y coordinates. Parush et al. [77] found out in an experiment in which the goal was to validate a complexity model that included alignment, grouping, and size variations that screens with grouping indications and alignment produced shorter search times than screens without grouping and poor alignment. A definition of UI alignment is provided by Vanderdonckt [97] as the proportion of objects that have a common straight line when their sides are extended. The alignment is guaranteed when the number of sides aligned, vertically and/or horizontally, is minimized.

B.1.1 Formula. Object alignment is equal to the proportion of objects that have a common straight line when their sides are extended. The alignment is guaranteed when the number of sides aligned both vertically and horizontally is minimized. Conversely, an interface having as many different alignment lines as the number of sides for each object is fully non-aligned.

External alignment. Several formulas for alignment are computable. It is possible to only take into account the coordinates describing the object as being its external points. Therefore, one possible formula to calculate alignment is:

$$A = \frac{DA_v + DA_h}{\sum_{i=1}^{n} C_i}$$

where DA_v and DA_h are respectively the number of segments aligned vertically and horizontally; C_i is the number of sides which are owned by object i; and n is the number of objects displayed. This formula takes only into account external alignment. It means that if we try to align two objects of different sizes, alignment on one of the external sides of each object will give the best measure.

Central alignment. Another possibility to assess the alignment of objects is to consider only the coordinates of their center. Then, one other possible formula to calculate alignment is:

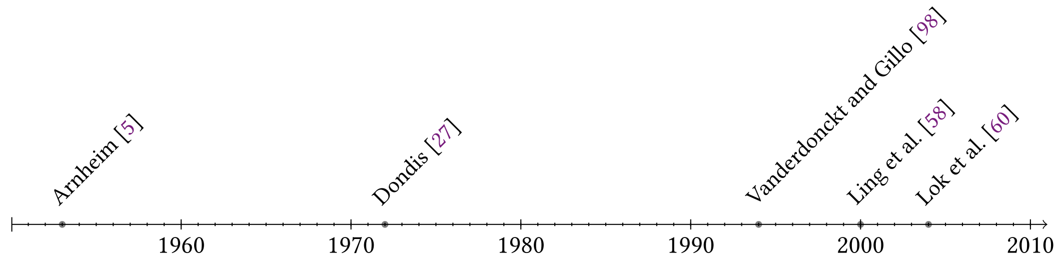
$$A = \frac{CAO_v + CAO_h}{n}$$

where CAO_v and CAO_h are respectively the number of center-aligned objects vertically and horizontally; and n is the number of objects displayed. This formula expresses central alignment.

B.2 Balance

Balance is a skill that everyone uses when walking in order to stand up and put a foot after another without falling. It is also a technique of compromise that is used in our everyday life (*e.g.*, find a balance between work and private life, time spent with friends and with a girlfriend)

Finding a good balance is also a crucial issue in every kind of design. In UI design, balance represents the extent to which a UI suggests some sense of equilibrium: how UI elements are arranged on the layout to reach some level of visual stability. It is often represented with the help of quadrants (*i.e.*, Upper-Left, Upper-Right, Bottom-Left and Bottom-Right) in order to determine which one is gathering a majority of UI elements and is, therefore, the "heaviest" quadrant causing a disequilibrium. Balance in art is often bound to symmetry but the truth is that there exist various ways to achieve balance in design and focusing on symmetrical balance is only one of them.



For ages, balance perception has certainly been a widely accepted principle in art techniques. Arnheim [5], a famous art theoretician, developed a whole chapter on this subject in his book "Art and Visual Perception: a psychology of the creative eye". Arnheim began the development around the concept of visual balance by presenting a plain black disk placed on a white square. He adds : "A yardstick would tell in inches the distances from the disk to the edges of the square. Thus it could be inferred that the disk lies off-center". Actually, he tries to focus the attention of the reader who will think: "I can see it at a glance, no need to measure that. . . " This is exactly the consequence of human's visual perception of balance. We do not need to measure exactly to see that something is slightly unstable or perfectly at equilibrium.

According to Dondis [27], the human need for balance has a strong influence on human perception. There is an analogy to the ability that man has to stand on his legs, "to have his two feet planted firmly on the ground and to know if he is to remain upright" : "no method of calculation is as fast, as accurate, as automatic as the intuitive sense of balance". Vanderdonckt and Gillo [98] introduces the concept of balance as being part of a catalog of visual techniques. It is defined as a search for horizontal and vertical equilibrium between existing interface objects. Ling et al. [58], guided by their goal to quantify aesthetics, introduces the optical weight in their definition of balance as the perception that one can have of an object weight. In a picture, the human eye perceives objects as more important than others.

The measure of balance then results from the difference between the total weight of the various components of each part of the vertical and horizontal axes, a mathematical formula. Lok et al. [60] introduce new categories of visual balance to dissociate symmetric, asymmetric, radial and crystallographic balance. Moreover, they implement the crystallographic balance by creating a WeightMap algorithm that consists in a bitmap of the visual weight at each pixel coordinate.

B.2.1 Formula. Balance as defined by Ngo and Byrne [71] is the distribution of optical weight in a picture. This optical weight is the perception that people can have of an object weight. Thus, in a picture, the human eye perceives objects more important than others. In terms of weight, we call these objects as "heavy". Dark colors, unusual shapes and large size add weight to objects. We can draw an analogy with a scale which is an instrument used by traders to determine the mass of commodities, and reason in the same way to determine the weight distribution within a UI.

Using the example below, we can easily determine that the first interface is perfectly balanced while the second presents a greater weight on the left side as it presents more objects, but also darker objects. The reasoning can be reproduced by comparing the upper and lower portions.

Balance is the difference between the total weights of the various components of each part of the vertical and horizontal axes. The formula is as follows.

$$BM = 1 - \frac{|BM_{vertical}| + |BM_{horizontal}|}{2} \in [0, 1]$$

$BM_{vertical}$ and $BM_{horizontal}$ are, respectively, the vertical and horizontal balance measures given by:

$$BM_{vertical} = 1 - \frac{w_L + w_R}{max(|w_L|, |w_R|)}$$

$$BM_{horizontal} = 1 - \frac{w_T + w_B}{max(|w_T|, |w_B|)}$$

where L, R, T, and B mean "left", "right", "top" and "bottom"; with

$$w_j = \sum_i^{n_j} d_{ij}(\frac{a_{ij}}{a_{max}} + |c_{ij} - c_{frame}| + s_{ij}) \quad (c_{ij}, c_{frame}, s_{ij}) \in [0, 1] \quad j = L, R, T, B$$

a_{ij}, c_{ij}, and s_{ij} are respectively areas, color and shape of the object i belonging to side j; a_{max} is the area of the interface object having the largest area; c_{frame} is the value given to the original color of the frame underlying the different objects; d_{ij} is the distance between the center of the object i and the segment nearest the frame of the side j; and n_j is the total number of objects belonging to side j. Every color is assigned a weight along a scale based on intensity *e.g.*, 0 (white) - 1 (black).

Shapes and colors. Regarding the form, we must refer to Birkhoff [13] formula providing a method for determining a measure for the shape of an object. It should be noted that we chose to study only the rectangular envelopes surrounding interface objects; the value for all object shape is then considered equivalent to a rectangle (1.25) and as all of them are similar, we could discard s_{ij} from the formula. Finally, and regarding color values, we chose to refer to the following framework to select the color intensity value.

Pixels-based formula. Another possibility to obtain a result for the measure of balance is to use a pixels-based formula. The principle is simple; we divide an interface in quadrants and count the number of full pixels and empty pixels in order to compute weights. Full pixels are filled with a color different from the frame, while empty pixels are those filled with a color equivalent to the frame. The changing part of the previous is the computation of quadrant weights given by:

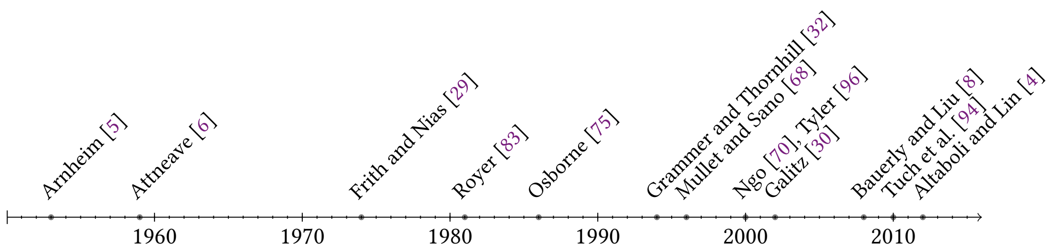
$$w_j = \frac{nPFull_j}{nP_j}$$

where $nPFull_j$ and nP_j are respectively the number of full pixels and the total number of pixels contained in quadrant j; a pixel is empty when its color is equal to color frame.

B.3 Symmetry

The concept of symmetry is very close to the one of balance. In Greek, συμμετρία , or *"symmetria"*, means agreement in dimensions, due proportion, and arrangement. Symmetry of a system is reached when some of its elements can be switched without losing its balance. Symmetry seems attractive because it induces an order that looks natural to humans. It is a very prevalent concept in our daily lives. Everything around us is symmetric because our body itself is symmetric and lots of objects we use every day are shaped to fit with our body.

A large portion of everyday objects is symmetric, ranging from the chair on which we sit to the car we drive. They are all designed with a symmetric shape in mind. Moreover, symmetry is not only static; it is also dynamic. Actions such as running, walking, swimming, and typewriting are all processed through a series of symmetrical steps. The most common example of symmetry associated with aesthetics that can be seen in the nature are the wings of a butterfly.



Arnheim [5] defines symmetry as the exact correspondence of form and constituent configurations on opposite sides of a dividing line or plane or about a center or axis. He considers that symmetry in a design creates centricity. Symmetric design must be used to focus the attention of the observer on the elements in its center. After all, symmetry is not only an art subject as human beings themselves have been designed symmetrically and there is empirical evidence that facial symmetry is a criterion of attractiveness between humans [32].

Frith and Nias [29] describe symmetry from a psychological point of view as *"constraint enforced on a stimulus to produce structure and redundancy"*. The stimulus being a set of elements in relation one with another and defined with various states. Attneave [6] argued well before that both beauty and information are in the eye of the beholder so that even a completely objectively unstructured stimulus might be qualified as structured by a human observer. Moreover, even when structure is present in multiple variations, the observer may be more focused on a few of these variations.

Therefore, people perceive a stimulus of vertical mirror symmetry more easily than of horizontal symmetry. Indeed, Royer [83] - who studied reaction times of symmetry detection - found that multiple symmetric patterns were faster to detect than single symmetric ones and that vertical symmetry was easier to detect than horizontal symmetry, itself easier to detect than diagonal symmetry.

According to Osborne [75], the modern conception is different from that inherited from classical times and from the Renaissance. Because our ideal of beauty in art is much more sensitive and emotional than rather directed by intellectual and mathematical principles as it was in the past, the representation of what is ugly finds also its place in pictorial art. Therefore a more general definition is accepted for symmetrical figures as "figures that are similar to each other in more than one way" [43].

Furthermore, by this acceptance of beauty of art as an expression and not reasoning, the author underlines the existence of aesthetic propositions which take the counterpart of the common accepted standards. Asymmetry appears then also as a property of aesthetics. Symmetry is an easy solution for designers to reach an attractive design by following simple and clear rules. It ensures balance and clear organization maybe sometimes at the expense of visual interest [68] and is therefore an easy technique for quickly building an efficient design. The degree of symmetry put forward by the designer would be ideally a fair compromise between the originality of an asymmetrical layout and the discipline of a symmetrical view. The aesthetic effect being easier to obtain with the later one, it can though lead to "unexciting" displays according to the authors.

As the prior goal of a UI is to be clear, appealing and efficient, symmetry is perfectly appropriate for its designing. Functional magnetic resonance imaging was used to explore how symmetry affected human brain [96]. It appeared that the representation of symmetry was identified as an activity specifically located in the occipital region.

Symmetry can be discriminated from noise in less than 100 msec. Ling et al. [58] are the first to define a measure for the symmetry quantification of a layout. The authors discriminate 3 properties of symmetry: vertical, horizontal, and radial. Bauerly and Liu [8] propose another formula for computing pixel-based symmetry based on comparisons of pixel pairs from both sides of a symmetry axis. Symmetry has lately been considered as a success factor for the crafting of appealing design. Tuch et al. [94] studied the effect of applied symmetry in website design on the user's aesthetic considerations. If their results corroborate the basic assumptions - namely the fact that symmetrically designed website has an impact on beauty appraisals while asymmetrically designed web pages were considered less attractive - they put forward another interesting aspect of symmetry perception which is a distinct consideration whether the subject is a man or a woman. It would therefore appear that men react more negatively to asymmetrical design than women do. This is an important result that should not be disregarded by designers.

B.3.1 Formula. Symmetry is the exact correspondence of form and constituent configurations on opposite sides of a dividing line or plane or about a center or axis, according to Arnheim [5]. Ling et al. [58] distinguishes three properties of symmetry: vertical, horizontal and radial and includes them into a formula that is:

$$SYM = 1 - \frac{|SYM_{vert.}| + |SYM_{hori.}| + |SYM_{radi.}|}{3} \in [0, 1]$$

$SYM_{vert.}$, $SYM_{hori.}$ and $SYM_{radi.}$ are the vertical, horizontal and radial symmetry with

$$SYM_{vert.} = \frac{\begin{array}{r} |X'_{UL} - X'_{UR}| + |X'_{LL} - X'_{LR}| + |Y'_{UL} - Y'_{UR}| + |Y'_{LL} - Y'_{LR}| \\ + |H'_{UL} - H'_{UR}| + |H'_{LL} - H'_{LR}| + |B'_{UL} - B'_{UR}| + |B'_{LL} - B'_{LR}| \\ + |\theta_{UL} - \theta'_{UR}| + |\theta'_{LL} - \theta'_{LR}| + |R'_{UL} - R'_{UR}| + |R'_{LL} - R'_{LR}| \end{array}}{12}$$

$$SYM_{hori.} = \frac{\begin{array}{r} |X'_{UL} - X'_{LL}| + |X'_{UR} - X'_{LR}| + |Y'_{UL} - Y'_{LL}| + |Y'_{UR} - Y'_{LR}| \\ + |H'_{UL} - H'_{LL}| + |H'_{UR} - H'_{LR}| + |B'_{UL} - B'_{LL}| + |B'_{UR} - B'_{LR}| \\ + |\theta_{UL} - \theta'_{LL}| + |\theta'_{UR} - \theta'_{LR}| + |R'_{UL} - R'_{LL}| + |R'_{UR} - R'_{LR}| \end{array}}{12}$$

$$SYM_{radi.} = \frac{\begin{array}{r} |X'_{UL} - X'_{LR}| + |X'_{UR} - X'_{LL}| + |Y'_{UL} - Y'_{LR}| + |Y'_{UR} - Y'_{LL}| \\ + |H'_{UL} - H'_{LR}| + |H'_{UR} - H'_{LL}| + |B'_{UL} - B'_{LR}| + |B'_{UR} - B'_{LL}| \\ + |\theta_{UL} - \theta'_{LR}| + |\theta'_{UR} - \theta'_{LL}| + |R'_{UL} - R'_{LR}| + |R'_{UR} - R'_{LL}| \end{array}}{12}$$

where X'_j, Y'_j, H'_j, B'_j, θ'_j and R'_j are respectively, the normalized values of X_j, Y_j, H_j, B_j, θ_j and R_j with Table 8 describing each of these variables and their formula where UL, UR, LL, and LR stand for upper-left, upper-right, lower-left, and lower-right, respectively; (x_{ij}, y_{ij}) and (x_c, y_c) are the co-ordinates of the centers of object i on quadrant j and the frame; b_{ij} and h_{ij} are the width and height of the object; and n_j is the total number of objects on the quadrant.

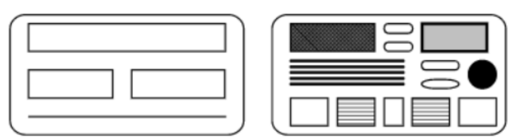
B.4 Density

Density of a UI may be defined as the part of the display dedicated to present information. In other words, and with some sort of abstraction, it is the ratio between the number of empty pixels and the number of full pixels.

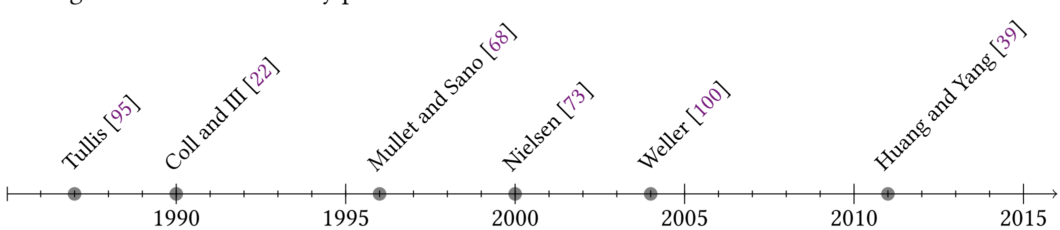
In a common sense, a clear interface with space between objects and no extra useless information looks easier to read than a dense one with lots of noise and no clear separation between objects.

Table 8. Decomposition of symmetry along dimensions.

Variable	**Description**	**Formula**
X	Symmetry on the x-axis	$\sum_i^{n_j} \lvert x_{ij} - x_c \rvert$
Y	Symmetry on the y-axis	$\sum_i^{n_j} \lvert y_{ij} - y_c \rvert$
H	Symmetry on objects height	$\sum_i^{n_j} h_{ij}$
B	Symmetry on objects width	$\sum_i^{n_j} b_{ij}$
θ	Symmetry on the ratio	$\sum_i^{n_j} \left\lvert \frac{y_{ij} - y_c}{x_{ij} - x_c} \right\rvert$
R	Symmetry on the Euclidean distance	$\sum_i^{n_j} \sqrt{(x_{ij} - x_c)^2 + (y_{ij} - y_c)^2}$



Using density with appropriateness gives to the user the impression that your UI is like a quill, such a light instrument that may produce a lot of information.



Tullis [95] defines a number of measures to consider when it comes to evaluating alphanumeric interfaces. Among them, he cites the overall density and the local density. The overall density is “the percentage of the display used to present information” while the local density is “the percentage of space used within each individual group of items”. In his model based on users’ performance and preferences, he found that search times were strongly influenced by density and grouping. Put another way and with a certain logic, it seems that users were more efficient in their visual search when the number of objects on the screen was low and when the objects of the same category were gathered together using extra space between those groups of objects.

Considering alphanumeric displays, Tullis [95] also includes in his studies mathematical formulas for overall density - expressed as ”the number of filled character spaces as a percentage of total spaces available”’ - and the local density - expressed as ”the average number of filled character spaces in a 5 deg visual angle around each character, as a percentage of available spaces in the circle and weighted by distance from the character".

Coll and III [22] conducted an experiment in order to investigate the effect of density on user preferences. Actually they were investigating complexity that could be reduced - according to them and with the context of contemporary designs in mind - to density. They asked 60 subjects separately to compare in terms of performance and appeal three screens with each one a specific density level (low, medium and high). The results showed that a screen appeal was following an inverted U-shaped curve regarding its density.

Later on, Huang and Yang [39] reproduced the same kind of experiment with 129 participants evaluating website homepages and with a focus on manipulating the density level (moderate or high). The findings of this last experiment indicate that participants prefer websites with a moderate-density homepage. White space is a full-fledged actor among design elements. It is used to describe a part of the layout that does not contain any text or graphics. It is important for improving readability and for the separation of content [68]. Using white space improves reading comprehension and could also enable to lead the user into a pre-defined visual search path [73].

White space can take many forms in visual design: line spacing, letter spacing, margins, padding, table cells size, etc. According to Ling et al. [58], density is the extent to which the screen is covered with objects and can be measured as the percentage of the entire frame covered by objects.

Weller [100] studied the effect of density on visual web search conducting an experiment where participants were instructed to find target words on a series of web pages. Each page varied in terms of contrast, local and overall density. Results indicated not only that search time was increasing with overall density but also that the high overall density was subjectively the least preferred screen presentation.

B.4.1 Formula. According to Ling et al. [58], density is the extent to which the screen is covered with objects and can be measured as the percentage of the entire frame covered by objects. Thus, inducing the following formula:

$$DM = 1 - \frac{\sum_{i}^{n} a_i}{a_{\text{frame}}}$$

where a_i and a_{frame} are the areas of object i and the frame; and n is the number of objects on the frame.

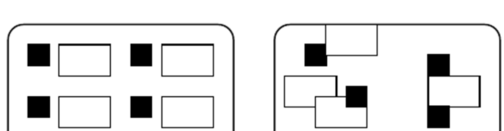
Pixels-based formula. In the same way, as it was computed for the measure of balance, it is possible to use a pixels-based formula counting the number of pixels contained in the UI and discriminating between full pixels and empty pixels. Full pixels are filled with a color different from the frame.

$$DM = \frac{nPFull}{nP}$$

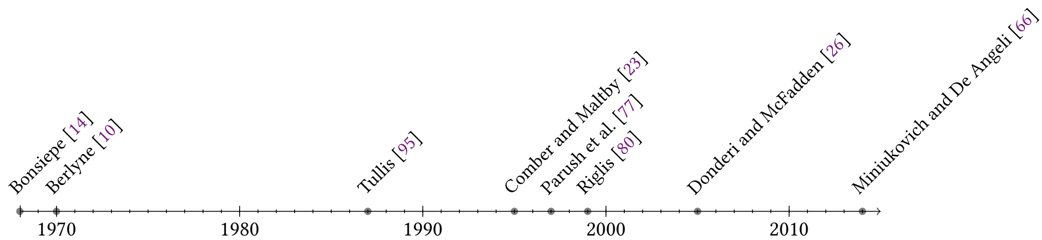
where $nPFull$ and nP are respectively the numbers of full pixels and the total number of pixels contained in quadrant j; a pixel is empty when its color is equal to the color frame.

B.5 Simplicity

Simplicity, best known in the literature as its antonym the visual complexity [94], describes how an interface is simple enough to be all at once understandable and sufficiently consistent.



According to Tullis [95], complexity is the degree to which the organization of the elements follows a predictable visual scheme.



Bonsiepe [14] considers the order of a system as a property impacting its aesthetics. A distinction is made between two types of order: (1) the order of the system where typographic items are regarded as elements of a system. This order is a function of the data needed to describe all the items occurring - the number of dimensional classes and the frequency of occurrence of the various dimensions and (2) the order of the arrangement based on the connections made between the items within an arrangement. It is a function of the number of different horizontal and vertical lines and the frequency with which the corners of items fall on these lines.

To determine the complexity of the system as an order, Bonsiepe used the Shannon formula (also known as the Boltzmann-Gibbs-Shannon entropy formula). In his theory of aesthetics, Berlyne [10] draws a relation between pleasure and stimulus. Specifically, he stated that this relationship follows an inverted U-shaped curve. Stimulus with moderate arousal is pleasant while low or high stimulus is unpleasant, one for being boring and the other irritating. An objective method for measuring stimulus complexity is by measuring the physical properties of an object, e.g. number of elements, dissimilarities and the degree of unity of several grouped elements. In his work on the evaluation of alphanumeric interfaces, Tullis [95] defines layout complexity as "the extent to which the arrangement of items on the frame follows a predictable visual scheme". He then derives a layout complexity measure as "the amount of information, in bits, conveyed by the horizontal and vertical positions of the display items". This measure is essentially based on the alignment of objects on the layout.

Basically, Tullis reused Bonsiepe's measure of complexity removing the system order part and keeping only the distribution order that is measured by counting the number of horizontal and vertical distances between alphanumeric items. Comber and Maltby [23] define a layout with a minimum complexity as "a layout where all objects are the same size and are aligned as neatly as possible to a grid. Maximum complexity occurs when every object has a different size and no two objects are aligned". In a pilot experiment, they studied the usability of 4 screen layouts designed with various complexity levels and assessed the complexity score with the help of Bonsiepe's measures. The results from this pilot study showed differences in usability between screens differing in complexity. They mostly observed that focusing on using a regular grid to minimize complexity was not the best way to layout a screen and to foster usability but leaving too much entropy in the design also negatively impacted usability. The goal for the designer is therefore to strike a balance by targeting medium complexity.

Parush et al. [77] puts forward in his research a model of GUI complexity based on four measurements of complexity: size complexity, the standard deviation of size, density, alignment, and grouping. Information obtained after compressing a file can be utilized as a reliable, valid, and objective measure of complexity.

Riglis [80] suggested measuring complexity through a ratio of bytes that are preserved after compression. Donderi and McFadden [26] conducted experiments that used compressed file length to predict search time and errors on visual displays used for marine electronic charts and radar similar to the one depicted.They found out that lossless JPEG file size was a good predictor of search performance and subjective complexity judgments. Miniukovich and De Angeli [66] found the existence of eight low-level determinants of GUI complexity. These determinants were categorized into classes related to the variation in information amount, organization and discriminability as follows: information amount, information organization and information discriminability. He defined pixel-based measures for the calculation of five of these determinants (color, visual clutter, contrast, edge congestion and symmetry) and tested the measures on 140 webpages rated beforehand by 10 participants on their aesthetics and complexity. The results indicate that a model of aesthetics including measures of color depth, dominant colors, edge congestion and visual complexity was highly predictive (R^2=0.57).

B.5.1 Formula. Simplicity is the directness and singleness of layout, free from secondary complications or sophistications [98]. The goal of simplicity is to reduce the necessary cognitive load to understand a UI. Simplicity, or its corollary complexity, depends on many different aspects visible on a layout such as variety of colors, shapes, positions, alignment and proportions. Basically the more variety in all aspects of a UI, the more complex it is. Therefore a possible formula is to combine different measures such as alignment, similarity and density into:

$$SIMP = \frac{s + a + 2d}{4}$$

where s, a and d are respectively measures for similarity, alignment and density given by:

$$s = \frac{n + 1 - n_{dis}}{n} \qquad a = \frac{2n - n_{ap}}{n} \qquad d = \frac{\sum_{i}^{n} a_i}{a_{frame}}$$

where n is the number of interface objects, or regions; n_{dis} is the number of dissimilar regions varying in proportion and color within a determined threshold; n_{ap} is the number of alignment points determined as different within a threshold; a_i and a_{frame} are respectively the area of object i and the frame.

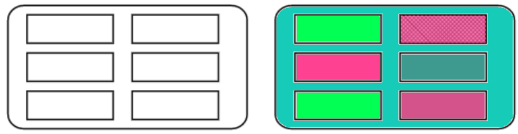
Image compression. Another possibility for computing complexity would be to use the size of a compressed image. It is intuitively acceptable as image compression algorithms operate as a pattern detector and proceed with compression in order to avoid repeating pixels that are close and similar. Therefore a simple UI, in the sense of a compression algorithm, would include few objects quite similar and orderly arranged while for a complex UI, it should be the complete opposite. The idea is to find a linear interpolation between the simplest image (blank) possible and the most complex one (random pixels). Therefore, we can use the following formula:

$$SIMP = \frac{fs_{UI} - fs_s}{fs_c - fs_s}$$

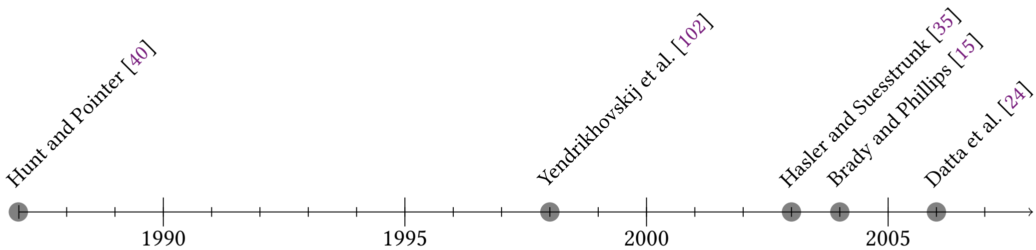
where fs_s, fs_c and fs_{UI} are respectively the file size of a blank image, the file size of a complex image - a blank image filled with color pixels randomly selected and arranged on the screen - and the file size of a screenshot of the evaluated UI. All images must have the same resolution in order to keep the measure consistent.

B.6 Colorfulness

Color is very important in our lives. It enables us to make distinctions between things, assess the harmfulness of our environment *e.g.*, when a metal looks orange/red, we can easily evaluate its heat and handle it carefully or when the sky is blue and clear, we know that we may go out without an umbrella without any risk.



Color in nature is also a very important means to convey a message in order to appeal or to repel. The human cortex is able to highlight color stimuli and has a dedicated area for color processing. The first measures for the computation of GUI colorfulness did not make it before the beginning of the 90s as it was linked to the arrival of color computers on the market.



Newton worked on the color spectrum (not indicated on the above timeline as it would have created a huge gap in the representation scale) as he undertook with the purpose of understanding color. His experiment consisted in drilling a very small hole in the wall and sending through this hole some natural light resulting in a light ray crossing the room to reach the opposite wall. Then, in a second step, he placed a prism of glass at the start of the ray and observed a separation of light into a spectrum of its component colors. Therefore proving that pure light was not fundamentally white or colorless. According to Hunt and Pointer [40], there exist 3 basic color attributes:

- *Brightness* is an "attribute of a visual perception according to which an area appears to exhibit more or less light".
- *Hue* is an "attribute of a visual perception according to which an area appears to be similar to one, or to proportions of two, of the perceived colors red, yellow, green, and blue".
- *Colorfulness* is an "attribute of a visual perception according to which an area appears to exhibit more or less of its hue".

Hunt and Pointer [40] propose various measures for color quantification, we retained 2 of them for the sake of argument:

- *Chromaticity* takes generally the form of a vector representing the relative amount of the 3 tristimulus values standing for the intensity of each RGB color.
- *Whiteness* indicates the degree of white contained in a color, it is a measure impacted by luminance and chromaticity.
- *Colorimetric purity* is a measure that correlates approximately with saturation but that is computed differently as it is a function of the luminance of the color stimulus and the luminance of the reference white that matches the color stimulus in an additive mixture.

With the purpose of optimizing color reproduction of natural images, Yendrikhovskij et al. [102] proposed a colorfulness index depending on two factors. The first factor is the distance of the image colors from a neutral gray that is estimated as the mean of the distribution of the chroma values of colors in the image. The second factor is the distance between individual image colors and is estimated by the standard deviation of the distribution of chroma values. The resulting measure replaces chroma by saturation and is given by:

$$C_i = S_i + \sigma_i$$

where S_i is the average saturation value of the image and σ_i is the standard deviation of saturation values distribution of the image. Hasler and Suesstrunk [35] proposed a very simple method for measuring colorfulness. It consists in using the very simple color space sRGB where each image pixel has a value between 0 and 255 for each color channel *i.e.*, Red, Green and Blue. Then by a combination of colors and with the computation of the mean and standard deviation, they obtained a measure that gives a score to each image in the sample dataset, the scores were correlated with the scores obtained through previous method at a degree of 95% which indicates that the measure is highly reliable. They split the scores into 7 classes of colorfulness attributes.

Brady and Phillips [15] explored in an experiment the impact of color manipulations of perceived usability. Participants clearly tended to perceive usability of an original website higher than the usability of its reproduction with altered colors.

Datta et al. [24] proposes another method to compute relative color distribution. He used the *Earth Mover's Distance* (EMD), which is "a measure of similarity between any two weighted distributions". Basically it is a type of color quantization that divides the RGB color space into 64 blocks. A first distribution of colors is generated as the color distribution of an image containing exactly the 64 colors. A second distribution of colors is computed from the original image as the frequency of occurrence of the 64 colors within the image. The EMD is then measured based on the pairwise distance between sampling points in the two distributions given by the Euclidean distances between the geometric centers of each cube.

B.6.1 Formula. Colorfulness measures the intensity of colors, or simply stated how colorful is an image. A measure for colorfulness is proposed by Hasler and Suesstrunk [35] using color space sRGB where each image pixel has a value between 0 and 255 for each color channel *i.e.*, Red, Green and Blue. The first step is to compute matrices from this RGB color space with:

$$rg = R - G \qquad yb = \frac{1}{2}(R + G) - B$$

In a second step, combined standard deviations and means of these matrices are calculated into a colorfulness measure with:

$$\sigma_{rgyb} = \sqrt{\sigma_{rg}^2 + \sigma_{yb}^2} \qquad \mu_{rgyb} = \sqrt{\mu_{rg}^2 + \mu_{yb}^2} \qquad M = \sigma_{rgyb} + 0.3\mu_{rgyb}$$

where R, G and B are the pixels color matrix having for each pixel a value corresponding relatively to red, green and blue channels; σ and μ are the symbols for respectively the standard deviation and the mean.

Color density. Extracting the colors with a color quantization algorithm generally produces a list of colors with, for each color, its individual numerical share in the image. These values can be used to compute a measure of density based on colors given by:

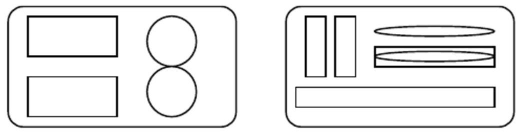
$$DM = 1 - \frac{c_{max}}{\sum_i^{n_c} c_i}$$

where c_i is the share value of color i in the image; c_{max} is the share value of the color that is the most represented; and n_c is the number of extracted colors.

B.7 Proportion

Since ages, human beings have shown that they have a preference for specific values determining the ratio between the height of an object and its width. The most famous one is probably the *Golden*

Ratio = $(1+\sqrt{5})/2 = 1.618$ that was used a lot in ancient art and, with probably a lower aesthetic impact, commonly said as being the measure ratio for ID and credit cards (ISO/CEI 7810).



In the following lines, we will investigate further the history of proportion measures. Marcus (1984) being the only two contributors we mention, we did not use a timeline. Euclid wrote a definition of the Golden Ratio in Euclid's Elements: "A straight line is said to have been cut in extreme and mean ratio when, as the whole line is to the greater segment, so is the greater to the lesser." A line AB is divided in an extreme and mean ratio by C if $AB/AC = AC : CB$. It is possible to find a position to cut a line in two segments where the ratio between the whole line and the greater segment is equal to the ratio between the greater segment and the smaller segment. This ratio has been shown to be equal to $(1+\sqrt{5})/2$.

Marcus [63] is a graphic designer with experience in information design and computer graphics and a faculty member of the University of California. In his research works on computer graphics, he strove for an aesthetically appealing ratio between the dimensions of interface objects. Because dimensions exist in the real world, we can feel them, we can see them, we can compare them. Mirroring dimensions in layouts consists of retracing this feeling, this illusion in the user interface. The ratio is calculated by dividing the height of an object by its width. Several proportions have been either proved aesthetic (*e.g.*, the Golden Ratio 1:618 used by the Greeks) or widely and conventionally preferred (e.g. the square - 1:1, the double square - 1:2, the root square of two, 1:1.414 and the root square of three: 1:1.732) as recommended by Marcus [63]. Disproportion - the opposite of proportion - is implied at the time no special ratio is used or a large difference appears between the two dimensions.

B.7.1 Formula. Proportion measures to what extent the interface objects are designed following some ratio considered as aesthetically pleasant e.g. squared, golden ratio, 4/9, $\sqrt{2}$, $\sqrt{3}$, etc.

According to Ling et al. [58], aesthetically pleasing proportions should be considered systematically when designing UI. He proposed a measure calculated as the relationship between dimensions of screen components and proportions with aesthetic predispositions given by

$$PM = \frac{|PM_{object}| + |PM_{layout}|}{2}$$

where PM_{object} and PM_{layout} are the difference between the proportions of respectively the objects and the layout with the closest aesthetics ratio given by

$$PM_{object} = \frac{\sum_i^n [1 - 2min(|p_j - p_i|]}{n} \qquad p_j = \frac{1}{1}, \frac{1}{1.414}, \frac{1}{1.618}, \frac{1}{1.732}, \frac{1}{2}$$

where n is the number of objects and p_i is the proportion of object i given by

$$p_i = \frac{min(h_i, b_i)}{max(h_i, b_i)}$$

where b_i and h_i are the width and height of object i. Note that the formula for computing PM_{layout} is a specification of the above-mentioned formula with the layout accounting alone for one object i and therefore where n is equal to 1.

C ALGORITHM PSEUDO-CODES

C.1 Algorithms for Computing Visual Measures

In order to facilitate the implementation of aesthetic measures, we have chosen to express each mathematical formula into a pseudo-code algorithm. A benefit of this process is to provide a basis for implementing the same measures in any other programming or markup language.

ALGORITHM 1: Balance

```
Function Balance (Layout Regions, Frame Center)
    Find weights for each side with:
    forall r ∈ LayoutRegions do
        forall j ∈ (Left, Right, Top, Bottom) do
            weight_j += distanceToCenter(j, r) * areaIn(j, r) / area_max
        end
    end
    Compute vertical and horizontal components
    return global balance
Function distanceToCenter (j, r)
    if r overlaps frame center then
        split r and conserve its center on side j
    end
    return distance between r center and frame center
Function areaIn (j, r)
    if r overlaps frame center then
        split r
    end
    return area in side j
```

ALGORITHM 2: Simplicity

```
Function Simplicity (Layout Regions)
    create vertical/horizontal alignment points array and dissimilar regions array
    forall r ∈ LayoutRegions do
        if r center is not aligned to any existing vertical/horizontal alignment point then
            add r center to vertical/horizontal alignment points array
        end
        if r center is not similar to any dissimilar regions then
            add r to dissimilar regions array
        end
    end
    s = (number of regions + 1 − number of dissimilar regions) / number of regions
    a = (2 * number of regions − number of AlignmentPoints) / number of regions
    d = ∑ regions area / layout area
    return simplicity = 1/4 s + 1/4 a + 1/2 d
```

ALGORITHM 3: External Alignment

```
Function External Alignment (Layout Regions)
    DAV = 0
    DAH = 0
    forall r1 ∈ LayoutRegions do
        forall r2 ∈ LayoutRegions do
            if r1 ≠ r2 then
                if r1 top is aligned to r2 top or bottom then
                    increment DAV by 1
                end
                if r1 bottom is aligned to r2 top or bottom then
                    increment DAV by 1
                end
                if r1 left is aligned to r2 left or right then
                    increment DAH by 1
                end
                if r1 right is aligned to r2 left or right then
                    increment DAH by 1
                end
            end
        end
    end
    return ExternalAlignment = (DAV + DAH) / 4n
```

ALGORITHM 4: Central Alignment

```
Function Central Alignment (Layout Regions)
    CAV = 0
    CAH = 0
    forall r1 ∈ LayoutRegions do
        forall r2 ∈ LayoutRegions do
            if r1 ≠ r2 then
                if r1 center is vertically aligned to r2 center then
                    increment DAV by 1
                end
                if r1 bottom is horizontally aligned to r2 center then
                    increment DAV by 1
                end
            end
        end
    end
    return ExternalAlignment = (CAV + CAH) / 2n
```

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ALGORITHM 5: Symmetry

Function *Symmetry (Layout Regions)*
  forall $i \in [Left, Right]$ **do**
    forall $j \in [Top, Bottom]$ **do**
      forall $r \in LayoutRegions$ **do**
        if $r \in quadrant_{i,j}$ **then**
          $X[i][j] += |r_{center-x} - frame_{width}/2|$
          $Y[i][j] += |r_{center-y} - frame_{height}/2|$
          $B[i][j] += r_{width}$
          $H[i][j] += r_{height}$
          if $r_{center} \neq Frame_{center}$ **then**
            $T[i][j] += \left|\frac{r_{center-y} - frame_{height}/2}{r_{center-x} - frame_{width}/2}\right|$
          end
          $R[i][j] += \sqrt{(r_{ctr-x} - frame_w/2)^2 + (r_{ctr-y} - frame_h/2)^2}$
        end
      end
    end
  end
  forall $M \in [X, Y, B, H, T, R]$ **do**
    Normalize M
    $Symmetry_V += \frac{|M[0][0] - M[1][0]| + |M[0][1] - M[1][1]|}{12}$
    $Symmetry_H += \frac{|M[0][0] - M[0][1]| + |M[1][0] - M[1][1]|}{12}$
    $Symmetry_R += \frac{|M[0][0] - M[1][1]| + |M[1][0] - M[0][1]|}{12}$
  end
  return $Symmetry = \frac{Symmetry_V + Symmetry_H + Symmetry_R}{3}$

ALGORITHM 6: Density

Function *Density (Layout Regions)*
  forall $r \in LayoutRegions$ **do**
    $area += r_{width} * r_{height}$
  end
  return $Density = \frac{area}{frame_{width} * frame_{height}}$

ALGORITHM 7: Concentricity

Function *Concentricity (Layout Regions)*

$frame_{diagonal} = \sqrt{(frame_{width}^2 + frame_{height}^2}$

forall $r \in LayoutRegions$ **do**

$distances+ = \sqrt{(r_{ctr-x} - frame_w/2)^2 + (r_{ctr-y} - frame_h/2)^2}$

end

return $Concentricity = \frac{2}{n} * \frac{distances}{frame_{diagonal}}$

ALGORITHM 8: Equilibrium

Function *Equilibrium (Layout Regions)*

forall $r \in LayoutRegions$ **do**

$X+ = |r_{area} * [r_{center-x} - frame_{width}/2]|$

$Y+ = \left|r_{area} * [r_{center-y} - frame_{height}/2]\right|$

$A+ = r_{area}$

end

$eqX = \frac{2X}{n * frame_{width} * A}$

$eqY = \frac{2Y}{n * frame_{height} * A}$

return $Equilibrium = 1 - \frac{eqX + eqY}{2}$

ALGORITHM 9: Regularity

Function *Regularity (Layout Regions)*

Initialize empty array of dissimilar regions

forall $r \in LayoutRegions$ **do**

if r *not similar to one region in array* **then**

add r to array

end

end

n_{dis} = dissimilar regions array size

return $Regularity = 1 - \frac{n_{dis}}{n}$

ALGORITHM 10: Colourfulness

Function *Colourfulness (Canvas)*

$i = 0, p = 0, rg = [], yb = []$

forall $pixel \in Canvas$ **do**

$red = pixel[i]$

$green = pixel[i+1]$

$blue = pixel[i+2]$

$rg[p] = |red - green|$

$$yb[p] = \left|\frac{red + green}{2} - blue\right|$$

$i+ = 4$

$p+ = 1$

end

$\sigma_{rgyb} = \sqrt{\sigma_{rg}^2 + \sigma yb^2}$

$\mu_{rgyb} = \sqrt{\mu_{rg}^2 + \mu yb^2}$

return $Colourfulness = \sigma_{rgyb} + 0.3\mu_{rgyb}$

ALGORITHM 11: Proportion

Function *Proportion (Layout Regions)*

$Total_{diff} = 0$

forall $r \in LayoutRegions$ **do**

$min = 0$

$$r_{proportion} = \frac{min(r_{height}, r_{width})}{max(r_{height}, r_{width})}$$

forall $j \in AestheticRatios$ **do**

$r_{diff} = |r_{proportion} - j|$

if $r_{diff} < min$ **then**

$min = r_{diff}$

end

end

$Total_{diff} + = 1 - 2min$

end

$$\textbf{return } Proportion = \frac{Total_{diff}}{n}$$